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Unintended responses to IT-enabled monitoring: The case of the electronic logging device mandate

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Abstract

Research shows that electronic monitoring technologies can reduce illicit agent (e.g., frontline worker) behavior along monitored dimensions such that principals (e.g., firms) benefit. Drawing on theory from criminology regarding the concept of offense displacement, we explain why the benefits from increased monitoring may be mitigated if agents subsequently increase other types of illicit acts not subject to greater monitoring. We investigate this possibility by examining how a mandate requiring millions of truck drivers to adopt electronic logging devices to record their working hours affected different safety behaviors and outcomes. Using data from millions of driver inspections and crash counts before and after the mandate and a differences-in-differences identification strategy, we find greater monitoring due to the mandate achieved its first-order effect of reducing the frequency that drivers were in non-compliance with work hour rules (the monitored dimension). However, drivers for small firms that were most affected by the mandate also appear to have increased their frequency of unsafe driving (e.g., speeding). We also find that crash counts for small firms did not fall relative to large firms, and may have increased. These results call into question whether increased electronic monitoring has improved safety in this industry.

KEYWORDS

agency theory, difference-in-differences, displacement, econometrics, electronic monitoring

1 | INTRODUCTION

Greater availability of information technologies that better monitor agents' (e.g., workers') behavior has changed how tasks are executed (Bernstein, 2017). Many activities that were previously performed with limited oversight are now monitored with substantial clarity by both firms and regulators. Consistent with predictions from agency theory (Eisenhardt, 1989), empirical findings generally show that monitoring improves agent behavior along the monitored dimensions, such as reduced absenteeism (Banerjee, Duflo, & Glennerster, 2008), less shirking (Nagin, Rebitzer, Sanders, & Taylor, 2002), increased compliance with

standardized processes (Staats, Dai, Hofmann, & Milkman, 2017), and less fraud (Pomeranz, 2015). There is also evidence of positive enforcement spillovers (Rincke & Traxler, 2011); for example, Pierce, Snow, and McAfee (2015) reported that the adoption of monitoring technology at a restaurant chain reduced employee theft and also increased productivity.

This manuscript seeks to extend understanding of how the use of information technologies to monitor agents can result in negative spillovers and unintended consequences. Known as *dysfunctional behavioral responses* in agency theory (Prendergast, 1999) or *offense displacement* in criminology (Guerette & Bowers, 2009),

increased monitoring of certain activities may result in undesired outcomes in other dimensions. We theorize that if agents can financially gain by committing multiple types of illicit behaviors, then increasing monitoring to mitigate one form of illicit behavior may result in displacement toward other illicit behaviors not subject to increased monitoring (Yang, 2008). We further theorize that displacement effects will be larger for agents who were more likely to engage in the type of illicit behavior targeted by electronic monitoring. Thus, while electronic monitoring may provide more information to firms, it does not necessarily improve performance.

We test our theory by examining how a government mandate requiring over three million large commercial trucks operating in the United States to be equipped with electronic logging devices (ELDs) affected truck drivers' safety behaviors and outcomes (Federal Register, 2015). An ELD is a technology that automatically records a driver's driving time and other aspects of hours-of-service (HOS, which are federal work rules governing truck drivers) compliance by capturing data from a vehicle's engine. ELDs were mandated to improve driver compliance with HOS rules by replacing paper logbooks that are easy to falsify (Levy, 2015) and, in doing so, reduce crashes by decreasing the number of fatigued drivers on the road (Federal Register, 2015). Truck drivers have an incentive to violate HOS rules because the industry's widespread use of piece-rate pay (Masten, 2009) can incentivize them to work more hours than legally permitted (Scott & Nyaga, 2019). However, violating HOS rules is not the only form of illicit behavior truckers can undertake to increase their productivity and, hence, earnings. They can also speed and drive aggressively (Miller, Golcic, & Fugate, 2018). Thus, requiring ELDs may result in offense displacement whereby drivers seek to make up for lost earnings from complying with HOS rules by increasing the extent they drive unsafely. As such, it is uncertain whether the ELD mandate achieved its stated goal of reducing crashes (Federal Register, 2015).

To examine agents' offense displacement behaviors due to increased monitoring, we leverage the fact that many small trucking companies (carriers) waited until just prior to the initial December 2017 deadline to adopt ELDs (Miller, Bolumole, & Schwieterman, 2020; Smith, 2017; Tita, 2017) and others delayed adopting ELDs until a second, stricter enforcement phase took effect in April 2018 (Lockridge, 2018). Thus, we leverage the exogenous shocks of the two enforcement regimes to proxy carriers' adoption of ELDs and, hence, an increase in the extent to which drivers' HOS compliance was electronically monitored. We then draw on research by the Owner-Operator Independent Drivers Association (OOIDA) that identified a set of carriers that had been using ELDs for several years

(OOIDA, 2016). Since these carriers already utilized ELDs, they were, practically speaking, unaffected by the mandate and therefore serve as a control group because their drivers did not see a change in the extent that their HOS compliance was monitored. We obtain data from millions of roadside inspections of drivers from January 1, 2017 (well before most small carriers started using ELDs (Miller et al., 2020) through September 1, 2018 (5 months after the mandate took full effect). The inspection data include whether (a) drivers were compliant with HOS regulations and, if not, the specific violations found, and (b) all unsafe driving violations (e.g., speeding). We combine the inspection data with crash data from the same time period, which includes identifiers for the carriers that were involved in the crashes and when and where the crashes occurred.

We report several findings. First, the mandate had its intended first-order effect: increased electronic monitoring of HOS regulations reduced HOS violations. For example, the ELD mandate resulted in a 43.0% reduction in HOS violations in carriers with 1 truck, and a 47.0% reduction in violations in carriers with 2–6 trucks. We also find evidence of offense displacement: for carriers with one truck, unsafe driving violations increased by 26.0%, and for carriers with two to six trucks, unsafe driving violations increased by 16.7%. We find no significant evidence of crash reductions; in fact, our best estimates indicate crashes for carriers most exposed to an increase in electronic monitoring increased.

This research makes several contributions. First, we contribute to the operations literature on factors that influence compliance (Gray, Anand, & Roth, 2015) and safety (Brown, Willis, & Prussia, 2000; de Koster, Stam, & Balk, 2011; Wiengarten, Fan, Lo, & Pagell, 2017). In particular, while research shows that monitoring directly affects worker behavior (e.g., Nagin et al., 2002; Olken, 2007; Staats et al., 2017), fewer studies document indirect effects or unintended consequences of monitoring (e.g., Pierce et al., 2015). This is understandable given the difficulty of gathering data on behaviors that agents try to obfuscate. Our study, based on rich data of observed illicit behaviors, makes an important contribution in the context of worker safety and compliance with regulatory rules. Second, we contribute to the literature on displacement effects (Guerette & Bowers, 2009) by studying this issue in an industrial setting due to an increase in electronic monitoring. In contrast, the majority of displacement studies focus on property crimes (e.g., car theft) (Gonzalez-Navarro, 2013) and violent crime (Jacob, Lefgren, & Moretti, 2007). Third, this work extends knowledge regarding how electronic monitoring affects safety in the motor carrier industry using a research design permitting more confidence in causal identification in that, unlike prior research that examined

consequences of carriers' voluntarily adoption of electronic monitoring technologies (Cantor, Corsi, & Grimm, 2009; Miller, Saldanha, Rungtusanatham, Knemeyer, & Goldsby, 2018)—which raises concerns about endogeneity (Ketokivi & McIntosh, 2017)—we rely on exogenous changes in regulatory regimes to identify causal effects. Importantly, our results run counter to previous results: monitoring technology can result in offense displacement that can offset gains in performance and safety. Fourth, our analysis provides unique insights into the effects of a technology-focused policy that impacted the flow of most goods shipped in the United States and affected millions of workers. Because of the substantial costs associated with the mandate (roughly \$2.5 billion annually), the potential impact on public safety (the Department of Transportation [DOT] estimated that 1,844 crashes, 562 injuries, and 26 deaths would be avoided (Federal Register, 2015), and the widespread impact on supply chains (Cassidy, 2018; Patrick, 2018), it is important to understand the safety benefits associated with the ELD policy intervention. The results call into question whether the mandate improved safety in the U.S. commercial trucking industry, and thus suggest that the policy should be revisited.

The remainder of this paper is structured in five sections. Section 2 summarizes the relevant literature. Section 3 sketches the logic for the hypothesized effects. Section 4 explains the empirical setting, describes the research design, and details variables. Section 5 presents the main analyses, explains the robustness tests, and discusses threats to causal identification. Section 6 discusses theoretical contributions, managerial and policy implications, limitations, and directions for future research.

2 | LITERATURE REVIEW

2.1 | Effects of electronic monitoring on agents' behaviors

A principal-agent (PA) relationship exists when one party (the principal) delegates tasks to another party (the agent) on the former's behalf (Jacobides & Croson, 2001). Two issues in PA relationships are that agents' goals are often misaligned with the principal's and there is information asymmetry favoring agents whereby agents know more about how they perform delegated tasks (Eisenhardt, 1989). Monitoring is central in agency theory because it reduces information asymmetry (Jacobides & Croson, 2001). Agents are often viewed as "rational cheaters" (Nagin et al., 2002, p. 855) in that they seek to balance their received benefits from committing actions forbidden by principals (e.g., illicit acts) with the perceived

costs of such actions, which are a function of the perceived certainty and severity of punishment (Kleven, Knudsen, Kreiner, Pedersen, & Saez, 2011). When an agents' benefits from committing illicit acts exceeds expected costs, agents will cheat (Becker, 1968). These behaviors can range from relatively benign transgressions, such as TSA agents taking longer rest breaks (Anteby & Chan, 2018), to illegal acts, such as mortgage brokers committing fraud (Nguyen & Pontell, 2010) and mechanics passing cars that should have failed emissions tests (Gino & Pierce, 2010). Higher levels of monitoring increase agents' costs for committing illicit acts because monitoring increases the likelihood that such behaviors are detected and punished (Di Tella & Schargrodsky, 2003).

Several studies show that exogenous increases in monitoring intensity reduce agents' propensity to commit illicit acts. Nagin et al. (2002) report that call center personnel who were placed in a treatment condition whereby they were likely to perceive a decline in monitoring were subsequently more likely to lie about their productivity (thus inflating their earnings), with this effect greatest for workers who were dissatisfied with their employer. Di Tella and Schargrodsky (2003) report that increased auditing intensity reduced bureaucratic corruption in the form of over-paying for commonly utilized hospital items. Banerjee et al. (2008) and Duflo, Hanna, and Ryan (2012) report that increased monitoring of nurse and school teacher attendance, respectively, reduced the number of absentee days. Pierce et al. (2015) report that restaurant workers' identifiable theft fell by 22% following the introduction of electronic monitoring and, concurrently, suggesting that monitoring results in positive spillovers because the restaurants' average revenue increased by nearly \$3,000 per week and servers' average tip percentage increased (an indication of improved service quality). Staats et al. (2017) report that hand sanitization rates increased for hospital workers following the introduction of electronic monitoring designed to improve hand hygiene. However, they suggest the effectiveness of this intervention waned with the passage of time and that, upon deactivation, hand hygiene levels declined below pre-monitoring levels.

Our study contributes to this literature by demonstrating how increased monitoring along one dimension may result in offense displacement toward other dimensions that can have undesirable consequences. This type of study has been difficult to undertake because it requires an expansive data set that is rich in variables characterizing many different dimensions of inter-related behaviors. Pierce et al. (2015, p. 2316) note that they could not explore offense displacement because they lacked data on potential offense displacement behaviors. The mandatory switchover to electronic monitoring in an

already heavily monitored sector of the economy affords a unique opportunity to explore the potential for offsetting behaviors in PA settings.

2.2 | Displacement of illicit behavior

Guerette and Bowers (2009, p. 1333) define displacement as the “relocation of a crime from one place, time, target, offense, tactic, or offender to another as a result of some crime-prevention initiative.”

Temporal displacement occurs when actors delay committing illicit activity x at location p until a time in the future ($t + 1$). For example, Vollaard (2017) reports that illegal discharges of oil from ocean vessels increases after sunset, which coincides with a decrease in enforcement activities. Spatial displacement, the most discussed displacement form in the criminology literature (Eck, 1993), occurs when, “offenders switch from targets in one location to targets in another location,” (Guerette & Bowers, 2009, p. 1333). For example, Gonzalez-Navarro (2013) finds that the introduction of Lojack in select Ford vehicles in some states in Mexico resulted in a decline in thefts for the select vehicles in the Lojack states, but thefts for these vehicles increased in non-Lojack states.

The academic literature on offense displacement in industrial settings is sparse, as this topic has been primarily examined in the context of property and violent crimes (Hesseling, 1994; Repetto, 1976), tax evasion (Lopez-Luzuriaga & Scartascini, 2019), and bureaucratic corruption (Olken, 2007). Yang (2008) provides an exception. His study examines how importers in the Philippines changed illicit strategies to avoid import duties following several changes in enforcement regimes. After the Philippines required import shipments from nine countries with a declared value greater than \$2,500 be subject to pre-shipment screening, there was an increase in the percentage of shipments valued between \$500 and \$2,500 from these countries vis-à-vis countries not subject to pre-shipment screening. Once this threshold was lowered to \$500, the percent of shipments valued between \$500 and \$2,500 declined for the pre-shipment screening countries relative to exempt countries, but simultaneously the percent of shipments destined for export processing zones (which were exempt from duties) increased for the pre-shipment screening countries relative to excluded countries.

2.3 | Electronic monitoring in the trucking industry

The remote nature of trucking operations, which makes direct supervision prohibitively costly, has made the

industry an important market for electronic monitoring technologies (Cantor et al., 2009; Cantor, Corsi, & Grimm, 2006; Hubbard, 2000). Of relevance is the body of work examining how the use of information technologies to monitor truck drivers affects safety. Cantor et al. (2009) report that carriers whose fleets had a greater percentage of trucks equipped with ELDs had fewer HOS violations and fewer crashes; they further report that carriers with more frequent HOS violations had more crashes, suggesting that HOS violations mediate the relationship between ELDs and crashes. Hickman and Hanowski (2011) report that unsafe driving behaviors (e.g., speeding, hard braking) decreased at two carriers that installed comprehensive camera systems that recorded drivers' in-cab operations. Miller, Saldanha, et al. (2018) report that carriers with greater capabilities of monitoring their truck drivers using different electronic technologies have higher levels of compliance with HOS rules. However, they suggest this relationship is sigmoidal (S-shaped). The marginal benefits of monitoring initially increase up to a “tipping point” after which they decline again, leading to a plateau in compliance. They attribute the plateau effect to the diminishing marginal deterrent effect of greater monitoring when detection probabilities are already high (e.g., moving from 20 to 25% probability of detection has a more pronounced deterrence effect than moving from 70 to 75% probability of detection).

This work builds on these studies in three ways. First, it provides the first large-scale evaluation of the first-order and second-order consequences of the ELD mandate, which extends prior studies that have examined how between-firm differences in the use of electronic monitoring technologies, including ELDs, affect safety. Second, our research design gives greater confidence in causal identification (Shaver, 2019) in that we rely on exogenous imposition of laws requiring ELDs to serve as a plausible proxy for ELD adoption by smaller carriers. In contrast, prior studies (Cantor et al., 2009; Miller, Saldanha, et al., 2018) rely on data from carriers that endogenously adopt ELDs, and these carriers may be fundamentally different in unobservable ways from carriers that choose not to adopt ELDs. Third, this is the first study in this space to suggest that electronic monitoring may result in negative outcomes in the form of more frequent unsafe driving behaviors or higher crash rates. This calls into question the cost-benefit calculations of studies that assume these devices reduce crash rates (Pitera, Boyle, & Goodchild, 2013).

3 | HYPOTHESES

We begin by sketching why, from an agency theory standpoint, an exogenous increase in monitoring using

information technologies designed to reduce agents' propensity to commit one type of illicit act will affect agents' behaviors along the targeted dimension. We frame our hypotheses in the context of electronic monitoring for two reasons. First, unlike monitoring by other members of an organization (e.g., supervisors), information technologies cannot be corrupted through agents' behaviors such as bribes and trading favors (Di Tella & Schargrodsky, 2003). This issue is important because one reason principals can have challenges with getting agents to change their behaviors in ways that principals desire is that other actors tasked with monitoring the agents are corrupt (Roy, 1952) or overly sympathetic ("leniency bias") (Prendergast, 1999). In contrast, data from information technologies have similarities to information generated by independent third parties (Kleven et al., 2011; Yang, 2008), which should be more reliable. Second, information technologies allow for continuous, as opposed to episodic, monitoring (Zuboff, 1988). Thus, relative to monitoring via other persons, monitoring through information technologies should have stronger effects on reducing agent misbehavior on the target dimension but, conversely, may result in more displacement in that, by more completely closing off one type of utility-generating agent misbehavior, monitoring technologies make the displacement behaviors more attractive.

One reason that the increased monitoring of agents using information technologies should reduce illicit acts on the monitored dimension is that, by increasing the frequency at which agents are observed, agents should perceive a greater cost from committing such actions because they are more likely to be found in violation (Kleven et al., 2011). For example, installing a theft monitoring system in a restaurant's point-of-sale software greatly increased the observability of serving staff's behaviors, including potential theft, and consequently, resulted in lower levels of theft by serving staff (Pierce et al., 2015). Likewise, reducing the frequency of monitoring has been found to increase worker fraud because workers have less fear that any given fraudulent act will be detected (Nagin et al., 2002).

A second reason that using information technologies to monitor agents can reduce agents' illicit acts along targeted dimensions is that information technologies can provide less equivocal information about agents' actions vis-à-vis monitoring by other means (Zuboff, 1988). For example, agents may have a hard time distorting data collected by information technologies compared to monitoring through self-reported paper documents. For example, Forbes, Lederman, and Tombe (2015) note that airlines appeared to "game" reporting the time of aircraft arrivals to the Federal Aviation Administration when arrival times were

reported manually by the airlines. Once aircraft arrival information was transmitted electronically by airplanes' computer systems, these distortions disappeared. By providing less equivocal information about agents' behaviors, heightened monitoring using information technologies increases the probability that an agents' misbehavior will be detected conditional on the agent being observed (Kleven et al., 2011). As explained by Pomeranz (2015), information about agents' actions that are more easily verified conditional on an audit should result in "preventative deterrence." This is because agents will perceive a higher cost from undesired actions given expected costs are the product of audit probability and the probability that an undesired action will be detected conditional on an audit occurring (Niehaus & Sukhtankar, 2013). Therefore, we posit:

H1A : Rational cheating will decrease with increased monitoring of agents.

We now focus on one factor that may shape the heterogeneity in agents' responses to increased monitoring on the target dimensions. Drawing on the literature concerning tax evasion (Kleven et al., 2011; Slemrod, Blumenthal, & Christian, 2001)—which also assumes rational cheaters (Allingham & Sandmo, 1972)—we expect that agents that had the greatest propensity to commit undesired behaviors prior to the increase in monitoring will be more affected by its imposition. The rationale for this prediction is rooted in theoretical arguments advanced by Pomeranz (2015) that activities for which illicit behavior is limited are unlikely to be affected by increased monitoring intensity. It follows that agents who have a lower probability of committing undesired behaviors will be less affected by increased monitoring. We suggest:

H1B : The decrease in rational cheating due to increased monitoring will be greater for agents who were more likely to undertake rational cheating behaviors prior to the increase in monitoring.

Our focus now shifts toward agents' potential for offense displacement due to increased monitoring. We begin by describing conditions that would be conducive for such effects to occur. Drawing on Yang (2008), three conditions necessary for offense displacement are: (a) there exist multiple types of distinct undesired behaviors that deliver utility (benefits less costs) for agents, (b) monitoring targets only a subset of these behaviors such that agents' perceptions of principals' heightened monitoring does not affect the probability that other undesired behaviors will be detected, and (c) the subset

of monitored behaviors deliver more marginal utility to agents than the non-monitored behaviors.

Offense displacement in the rational cheater framework can be understood through the lens of the multi-tasking agent from agency theory (Holmström & Milgrom, 1994). Agents will respond to one form of undesired behavior becoming more costly by substituting a different form of behavior that will deliver them utility. Yang's (2008) examples of importers substituting different tactics to avoid import duties illustrates this process. Another example comes from the trucking industry, where drivers can respond to reduced work hours by increasing the pace of work (Ouellet, 1994), which can result in various unsafe behaviors behind the wheel. For example, drivers can exceed the speed limit, drive more aggressively (e.g., following too closely, sudden lane changes), and elect to communicate using cell phones while driving vis-à-vis pulling over for such communications as legally required.

Building on the logic underlying H_{1A} and H_{1B} , we expect offense displacement to be more pronounced for agents who recently saw an increase in electronic monitoring to curtail one type of undesired behavior due to agents' efforts to make up for diminished utility (Olken, 2007). We would further expect offense displacement to increase as monitoring becomes more intense because agents will exhibit a larger decrease in targeted behaviors, which will concomitantly increase the attractiveness of other behaviors to make up for lost utility. We would also expect offense displacement to be more pronounced for agents who were more likely to commit the now-monitored offense before the monitoring program was implemented, as these agents would see a larger decline in utility due to enhanced monitoring efforts (Pierce et al., 2015). We therefore suggest:

H2A : Offense displacement will increase with increased electronic monitoring of agents.

H2B : The increase in offense displacement will be greater for agents who were more likely to commit the illicit act that became subject to more intense monitoring.

When examining the effect of monitoring on principals' outcomes (e.g., productivity, sales, profits, accidents, etc.) a distinction can be made between outcomes that are directly related (i.e., proximal) to the monitored behavior versus those that are indirectly related (i.e., distal) to the monitored behavior. To illustrate, if a manufacturing plant increased monitoring of workers' adherence to proper use of protective equipment, plant-

level injury rates would be a proximal outcome while plant-level productivity would be a distal outcome (Christian, Bradley, Wallace, & Burke, 2009). This research concerns proximal outcomes only. Unlike the other hypotheses, it is infeasible to devise a generalized prediction regarding how increased monitoring of one behavior will affect principals' proximal outcomes due to the need to incorporate context-specific factors. Four factors of special importance are (a) the extent targeted undesired behaviors decline due to increased monitoring, (b) the marginal effect of the targeted undesired behaviors on the proximal outcome, (c) the magnitude of the offense displacement to non-monitored behaviors, and (d) the marginal effect of the offense displacement behaviors on the proximal outcome (Pierce et al., 2015; Yang, 2008; Zuboff, 1988).

In our research context, we expect the increased monitoring of truckers' compliance with HOS rules will, opposite arguments from the DOT (Federal Register, 2015), result in more crashes. One reason of this prediction is that fatigue, which is not necessarily tightly coupled with compliance to HOS rules (Ouellet, 1994), has a limited relationship with crashes. Craft (2010) reports that less than 2% of fatal large truck crashes are caused by drivers being drowsy, asleep, or fatigued. Thus, to the extent that heightened monitoring of truckers' HOS compliance reduces their fatigue levels, the marginal effect of this fatigue reduction on accidents is likely limited. The second reason for this prediction is that greater monitoring of drivers' HOS compliance using ELDs increased truckers' propensity to rush to complete tasks, with Viscelli (2018, p. 1) stating, "I am hearing about drivers rushing to get loads done when they wouldn't have before...[ELD implementation] prevents some abusive practices and long days, but it also forces drivers to rush and cut corners." Critically, behaviors of this sort have a greater marginal impact on crashes. Craft (2010) reports that unsafe driving behaviors have a stronger correlation with accident rates than fatigue, with speeding and failure to remain in the proper lane being the two most significant factors contributing to fatal large truck crashes. Our context-specific prediction aligns with Yang's (2008) more general observation that if targeted undesired behaviors are loosely linked to an outcome and increased monitoring results in offense displacement behaviors that have a larger negative marginal effect on an outcome, then enhanced monitoring may not make a principal better off. We therefore suggest:

H3 : Increased monitoring of truckers' hours-of-service rules using electronic logging devices following the ELD mandate increased crash rates.

4 | DATA AND VARIABLES

4.1 | Industry setting, HOS regulations, and the ELD mandate

To investigate these hypotheses, we use data from the trucking industry. Trucking is the dominant mode of transportation in the United States, moving roughly 67% of all tonnage nationwide (Corridore, 2014). The hundreds of thousands of for-hire carriers are a particularly important segment because they move goods through virtually every major supply chain (Caplice, 2007). Carriers range from independent owner-operators who own one truck (Monaco & Redmon, 2012) and small carriers with a handful of vehicles to very large carriers with thousands of trucks (Scott, Parker, & Craighead, 2017).

Due to carriers' impact on the motoring public, the DOT regulates trucks involved in interstate commerce via the Federal Motor Carrier Safety Administration (FMCSA), whose primary goal is to "prevent commercial motor vehicle-related fatalities and injuries" (FMCSA, 2018a, p. 1). The FMCSA regulates the legal working conditions of vehicles and drivers. Vehicles must be maintained in an acceptable condition; for example, they must have working headlights, cargo must be properly secured, and they must not exceed specified weight limits. Drivers must comply with HOS restrictions, which require rest breaks through the course of a workday and limit the total driving time in a day and over the course of several days to reduce driver fatigue (Stern et al., 2018).

Historically, the industry has been plagued by chronic non-compliance with HOS rules due to several factors (Chen et al., 2015), including (a) the predominance of piece-rate pay (Masten, 2009) that can incentivize drivers to work more hours to increase earnings (Scott & Nyaga, 2019), (b) intense intra-industry rivalry due to low entry barriers, especially for smaller carriers (Belzer, 2000), (c) low probability of inspection¹ due to the industry's size coupled with limited enforcement resources (Miller et al., 2020), and (d) the paper logbooks drivers used to record their work hours were easy to falsify (Viscelli, 2016). Due to this latter issue, the FMCSA passed rule 49 CFR Parts 385, 386, 390, and 395 (Federal Register, 2015) (henceforth, the ELD mandate) on

December 16th, 2015, which required nearly all commercial vehicles to use ELDs to track a driver's work hours by December 18th, 2017, and was phased in over two implementation periods. Prior to December 18, 2017, a driver could use a paper log ("pretreatment" period). After and including December 18, 2017, the FMCSA required drivers to use an ELD but enforced the mandate relatively lightly, only issuing discretionary citations for non-compliant drivers ("light enforcement" period). On April 1, 2018, the FMCSA started strictly enforcing the ELD mandate ("strict enforcement" period); for example, they ordered non-compliant drivers "out of service" for 10 hr if they did not have an appropriate ELD (Jaillet, 2017). Figure 1 provides a timeline.

The challenge facing the FMCSA is that while the ELD mandate should reduce noncompliance with HOS rules because it forced many carriers to adopt these devices (Miller et al., 2020), the mandate may have resulted in offense displacement whereby drivers increased their pace of work and the extent that they drove aggressively (Viscelli, 2018). As such, it is an empirical question as to whether the ELD mandate improved motor carrier safety and, consequently, the safety of the motoring public.

4.2 | Research design and causal identification

We are interested in studying how electronic monitoring affects the propensity of drivers to (a) violate HOS regulations, (b) commit unsafe driving violations, and (c) be involved in crashes. Ideally, we would randomly assign drivers to treatment and control groups, and then compare outcomes between these groups. While we cannot do this, the FMCSA can exogenously affect electronic monitoring by requiring interstate motor carriers to adopt ELDs. As noted in several sources (Miller et al., 2020; Smith, 2017; Tita, 2017), a large percentage of small- and mid-sized carriers did not adopt ELDs until just prior to light enforcement and vehemently opposed ELDs for years (e.g., Johnston, Cullen, Mayers, & Cullen, 2014). For example, Miller et al. (2020) reports that as of mid-September 2017 only 15% of motor carriers with 5–99 power units had started the process of adopting ELDs.

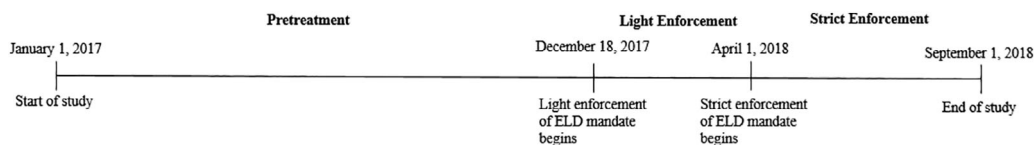


FIGURE 1 Timeline of key dates

Other industry research indicates that strict enforcement further increased ELD adoption. For example, Lockridge (2018) notes that, as of January 2018, a substantial number (e.g., 20%) of smaller fleets were still not compliant, with compliance increasing as strict enforcement loomed (Dorf, 2018). As it is unlikely most non-users would have adopted ELDs without the mandate—ELDs had been on the market for many years (Cantor et al., 2009)—smaller carriers are “treated” by the mandate’s two enforcement stages. Drawing on OOIDA (2016), we can identify a control group of carriers whose level of electronic monitoring was unaffected by the mandate because they already used ELDs. Thus, we can use a differences-in-differences (DiD) approach to compare the change in outcomes for the treated group before and after the mandate to the change in outcomes for the control group before and after the mandate. By subtracting the control group’s trend from the treatment group’s trend, we are left with a causal estimate of the effect of electronic monitoring on the carriers that were induced to adopt electronic monitoring by the ELD mandate since the control group’s trend accounts for other potential confounding factors.

While OOIDA (2016) provides us with an unequivocal control group, it is important to note that we cannot directly observe ELD adoption by other carriers to ensure all carriers in the treatment groups went from not having ELDs before the mandate to having them afterwards. However, this can only result in the *under-estimation* of treatment effects (Angrist & Pischke, 2010) because adopting carriers’ safety data are pooled with data from carriers that did not change their electronic monitoring levels.² The fact that treatment assignment is imperfect does not invalidate the use of a DiD design because, as Angrist and Pischke (2010, p. 14) note, “differences-in-differences policy analysis typically compares the evolution of outcomes in groups affected *more and less* by a policy change” [emphasis added]. This is akin to DiD studies in the tax evasion literature (Kleven et al., 2011) in that scholars cannot measure whether subjects placed in higher audit probability treatment groups are actually affected by the treatment in that they may comply with tax laws regardless of audit probability. Such instances also result in under-estimated treatment effects. Consequently, our research design results in conservative tests.

4.3 | Data sources

Our data come from the FMCSA, which collects data³ on inspections, crashes, and legally authorized carriers in order to monitor carrier safety. We use four sets of files for our analysis: (a) inspection files, (b) violation files,

(c) crash files, and (d) census files. We received the files on November 5 and 11, 2018, with data that are accurate through September 1, 2018. Our analysis focuses on the short-run impacts of the ELD mandate, and therefore uses data shortly before and after the mandate. As shown in Figure 1, the sample period begins on January 1, 2017 and ends on September 1, 2018. We focus our analysis on the carriers most affected by HOS regulations, namely, for-hire interstate carriers, excluding buses and private carriers from the sample (Miller et al., 2020; Scott & Nyaga, 2019).

The inspection file contains detailed information for every inspection performed by an authorized state or federal inspector during the sample period. DOT-approved inspectors conduct an average of more than two million roadside inspections per year across all 50 states and the District of Columbia. Roadside inspections are standardized into eight different categories. Depending on the category of roadside inspection, an inspector may check a driver’s documents—medical certificates, licenses, and electronic or paper logs—perform a cursory vehicle walk-around, or a more intensive vehicle inspection. The decision to inspect can be random, based on the FMCSA Inspection Selection System (ISS), or triggered at the discretion of the inspector or law enforcement officer because of moving violations or obvious issues of non-compliance. The inspection file includes when and where the inspection occurred, the type of inspection, which carrier was inspected (every legal carrier has a unique identifying number, called a DOT number), and the number of violations that were found during the inspection.

We are also interested in specific violations (e.g., driving longer than the legal time, skipping required breaks) listed in the violation file, which we link to the inspections via a unique inspection identifier. This file includes the exact violations reported by the inspector. Importantly, the violations file includes moving violations (such as citations for unsafe driving, or speeding). These citations and violations have a unique inspection number, but the causal order of inspection and violation is reversed to those of inspections triggered by ISS: the violation triggers an inspection, rather than an inspection uncovering a violation.

The crash file contains data on every crash involving a large truck or bus that resulted in an injury, death, or required a vehicle to be towed (Stern et al., 2018; called “federally recordable accidents”). This file also includes details such as when and where the crash occurred and, key for our purposes, the DOT number of the carrier that was involved.

Finally, the census files catalog every commercial carrier legally authorized to operate on federal roads and highways, including the carrier’s DOT number, whether they are an interstate and/or for-hire carrier, and the number of trucks they own. We use several census files—

one each from December 2016, October 2017, June 2018, and August 2018. To link a carrier's registration data, we use the closest census *before* a crash because a crash could affect a carrier—for example, whether they stay in business or the number of trucks they operate. Thus, the carrier census information will not be affected by any of the crashes in our sample time period.

In our sample, there were 4.0 million inspections performed by 15,266 inspectors on 224,878 for-hire interstate carriers. 191,576 (4.8%) of the inspections had at least one intentional HOS violation (intentional violations are violations such as falsifying a logbook, or not having a logbook), there were 268,427 unsafe driving violations, and 189,406 (70.6%) of those were speeding violations. There were 156,072 crashes involving a for-hire interstate truck during our sample period. For this paper, we refer to intentional HOS violations simply as HOS violations, ignoring paperwork errors.

4.4 | Variable construction

Our unit of analysis is the size cohort-week in that we categorize carriers according to size based on the total number of power units reported by the carrier and then aggregate all inspections performed by all carriers within a size cohort during a week into week-level aggregates. This strategy is essential because a small motor carrier experiences very few inspections, which makes estimation at the individual carrier level infeasible. In contrast, constructing size cohorts resolves this issue, as noted by Baker and Hubbard (2004). Furthermore, by aggregating

a large number of inspections across similarly sized carriers, random measurement error cancels out (Grunfeld & Griliches, 1960), which allows for the more precise estimation of treatment effects.

The size categories are defined as carriers with (a) 1 truck, (b) 2–6 trucks, (c) 7–20 trucks, (d) 21–100 trucks, (e) 101–1,000 trucks, (f) 1,001–50,000 trucks, excluding the carriers forming the control group, (g) more than 50,000 trucks, and (h) a set of carriers forming the control group. In our analysis, we exclude the category of carriers with 50,000 or more trucks (this essentially excludes UPS and FedEx from our analysis). These size categories are consistent with those used by the DOT when reporting truck and bus facts (FMCSA, 2018b), with three more categories to allow for a more detailed analysis. For the control group, we select a set of eight large asset-based motor carriers⁴ that operated using ELDs for many years prior to the ELD mandate (OOIDA, 2016). Because they operated ELDs well before the mandate, they should be relatively unaffected by a mandate requiring the use of ELDs. The control group averaged over 71,000 power units in operation during the time period we investigate, which creates a large control group⁵ that is subject to an average of 1,215 weekly inspections during our study period. With weekly aggregation, there are 87 weeks of observations for each of the seven size cohorts, resulting in $NT = 609$ observations in total. As shown in Figure 1, Week 1 starts on Sunday, January 1, 2017, and Week 87 ends on Saturday, September 1, 2018. The light enforcement period corresponds to Weeks 51 through 65, and the strict enforcement period corresponds to Weeks 66 through 87. Table 1 summarizes the size cohort

TABLE 1 Carrier size categories and descriptive statistics for each category

Carrier-size category	Number of carriers	Inspections (% with an HOS violation)	Unsafe driving violations	Crashes	Included?
Trucks not reported	945	5,189 (3.5%)	258	622	No
1 Truck	108,698	416,202 (8.8%)	26,317	13,760	Yes
2–6 Trucks	81,564	698,750 (7.5%)	44,944	21,529	Yes
7–20 Trucks	30,446	723,548 (5.6%)	46,426	22,684	Yes
21–100 Trucks	13,271	977,999 (4.1%)	63,935	32,901	Yes
101–1,000 Trucks	2,553	708,912 (2.3%)	50,659	33,191	Yes
1,001–50,000 Trucks ^a	191	355,202 (1.3%)	24,651	21,612	Yes
More than 50,000 trucks	9	18,391 (0.7%)	2,591	3,452	No
Control group	8	105,710 (0.8%)	8,646	6,321	Yes
Total	237,685	4,009,903 (4.8%)	268,427	156,072	

Note: There are 9 carriers with more than 50,000 trucks, but 7 of those appear to have reported incorrect information to the DOT. For example, UPS and Fedex account for more than 99.9% of the inspections in that size category. 945 carriers did not report the number of trucks they operate. 97.4% of all crashes, 98.9% of all unsafe driving violations, and 99.4% of all inspections are included in the analysis.

^aThis group excludes large asset-based carriers identified by OOIDA (2016).

TABLE 2 Descriptive statistics by treatment period

Carrier-size category	Pretreatment			Light enforcement			Strict enforcement		
	Inspections	Unsafe driving violations	Crashes	Inspections	Unsafe driving violations	Crashes	Inspections	Unsafe driving violations	Crashes
1 Truck	223,210	13,404	7,737	75,148	4,934	2,501	117,844	7,979	3,522
2–6 Trucks	403,287	24,577	12,115	115,631	7,660	3,988	179,832	12,707	5,426
7–20 Trucks	420,345	25,818	12,703	119,417	7,983	4,237	183,786	12,625	5,744
21–100 Trucks	565,507	35,246	18,683	163,048	10,972	6,092	249,444	17,717	8,126
101–1,000 Trucks	406,393	28,667	18,717	118,759	8,592	6,521	183,760	13,400	7,953
1,001–50,000 Trucks ^a	198,766	13,755	12,233	61,242	4,108	4,090	95,194	6,788	5,289
Control group	60,657	4,917	3,668	19,071	1,447	1,245	25,982	2,282	1,408
Total	2,278,165	146,384	85,856	672,316	45,696	28,674	1,035,842	73,498	37,468

^aThis group excludes large asset-based carriers identified by OOIDA (2016).

descriptive statistics and Table 2 shows these statistics by treatment period.

Rational cheating is operationalized as the rate of HOS violations. $HOS\ Violation\ Rate_{it}$ is the percentage of inspections for carriers in size cohort i during week t that had an HOS violation. Offense displacement is operationalized by the frequency of unsafe driving behaviors. $Unsafe\ Driving_{it}$ is a count of the number of unsafe driving violations involving carriers in size cohort i during week t . Unsafe driving violations include speeding, failing to obey traffic signals, following too closely, and improper lane changes. In order to examine percentage changes, we take the natural logarithm of these weekly counts, which we denote as $Log\ Unsafe\ Driving$. Principals' proximal outcomes are captured by crashes. $Crashes_{it}$ is a count of the number of federally recordable crashes involving carriers in size cohort i during week t ; we also use the logarithmic transform, which we denote as $Log\ Crashes_{it}$.

4.5 | Evaluating DiD assumptions

The veracity of our causal inference rests on several assumptions. First, the “treatment” event must be exogenous of other factors that could affect carriers' safety. This assumption holds because the ELD mandate was implemented industry-wide by the FMCSA, meaning that adoption was not voluntary. Second, it is important that inspectors' behaviors remain unchanged before and after the mandate. To examine the potential for this sort of bias, we look at both aggregate and relative trends in inspections. In Figure 2a we plot the number of total inspections for the 35 weeks that are common to both 2017 and 2018. There is no apparent deviation between

the patterns of aggregate inspections in 2018 relative to 2017. In Figure 2b we plot the number of weekly inspections by each size cohort; in general, the lines are parallel, though there appears to be some uptick in the number of inspections for the 1 truck cohort. Consequently, we include an additional control variable labeled $Inspection\ Proportion_{it}$ that represents the proportion of total inspections that were conducted on size cohort i for week t . Since total inspections remained unchanged, $Inspection\ Proportion$ will control for changes in either (a) the propensity to be inspected or (b) on-road fleet mix, or both.

A third critical assumption is that the control group acts as a valid counterfactual—that in the absence of treatment, the treatment group *would have* followed the same trend as the control group, known as the “parallel trends” assumption (Angrist & Pischke, 2009). While this assumption involves a counterfactual and is therefore impossible to test explicitly (Holland, 1986; Imbens & Wooldridge, 2009), we can observe whether treatment and control groups moved in parallel *prior* to the mandate. We evaluate this graphically and statistically. Plots of the weekly data for the control group for common weeks and split across the size cohorts for the full sample period for $HOS\ Violation\ Rate$, $Log\ Unsafe\ Driving$ and $Log\ Crashes$ are shown in Figures 3a,b, 4a, b, and 5a,b. Visual inspection suggests that there was no practical change in these measures for the control group before and after the mandate, and that the parallel trend assumption holds for $HOS\ Violation\ Rate$ and $Log\ Unsafe\ Driving$; it may be less tenable for $Log\ Crashes$. Statistically, we fit two sets of panel data regression models using the Parks estimator (Parks, 1967), whereby we estimate the linear trend for

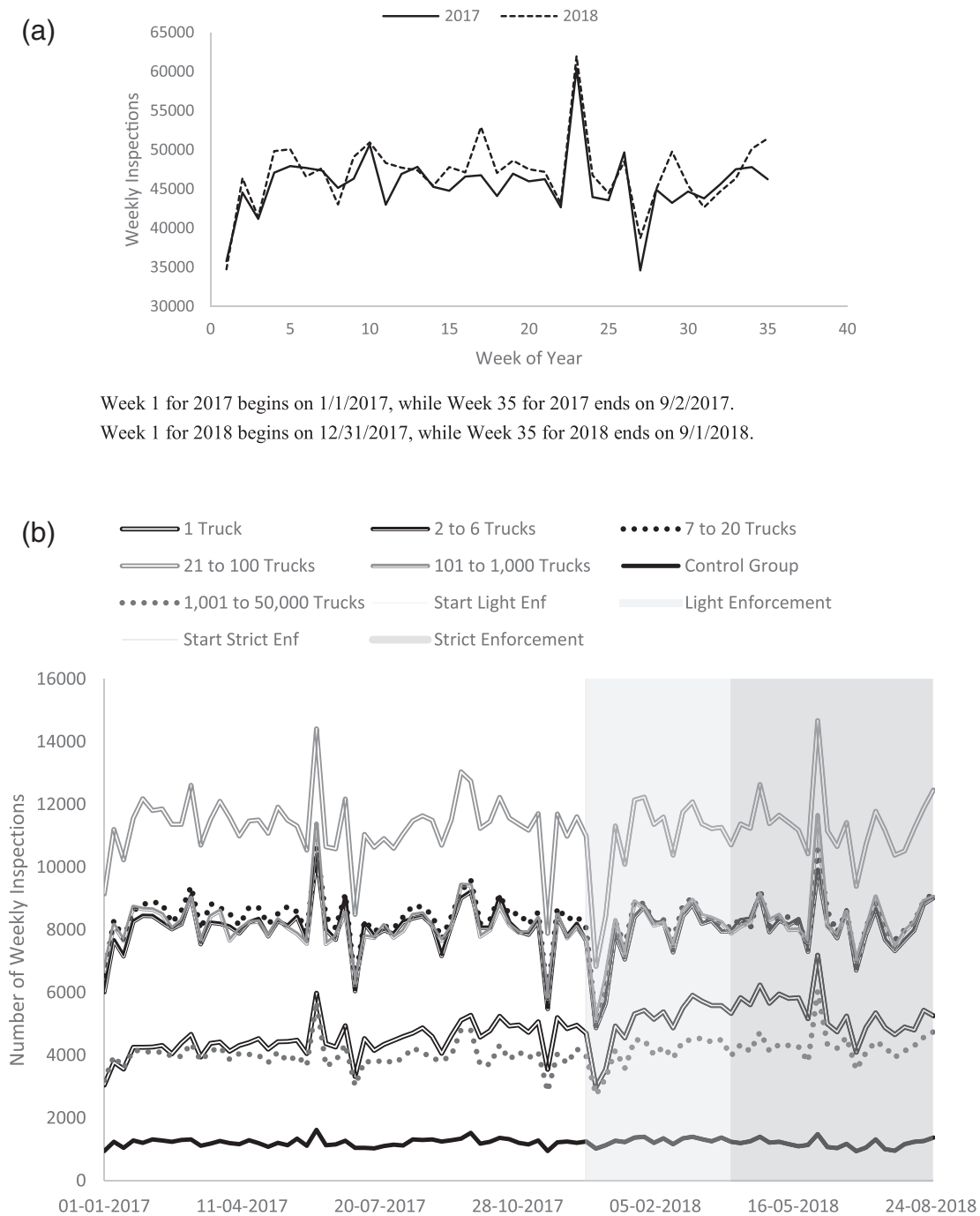


FIGURE 2 (a) Weekly inspections for the weeks common to 2017 and 2018. Week 1 for 2017 begins on 1/1/2017, while Week 35 for 2017 ends on 9/2/2017. Week 1 for 2018 begins on 12/31/2017, while Week 35 for 2018 ends on 9/1/2018. (b) Weekly inspections by size cohort

each size cohort relative to our control group through (a) August 2017 (Week 35) and (b) the week prior to light enforcement of the ELD mandate (Week 50). We report these results in Table 3.

Starting first with Table 3 Panel A for *HOS Violation Rate*, we see that, through August 2017 (Model 1), all size cohorts except the 101–1,000 truck group displayed similar trends, consistent with the discussion in

§4.2 that, while many carriers waited until just before the ELD mandate to adopt, others began adopting ELDs in anticipation of the mandate. Model 2's results are consistent with anticipation, as the 2–6 truck and 21–100 truck cohorts begin to trend down slightly relative to the control group. However, as is clearly visible in Figure 2b, the downward trend cannot account for the discontinuous decline in *HOS Violation Rate*,

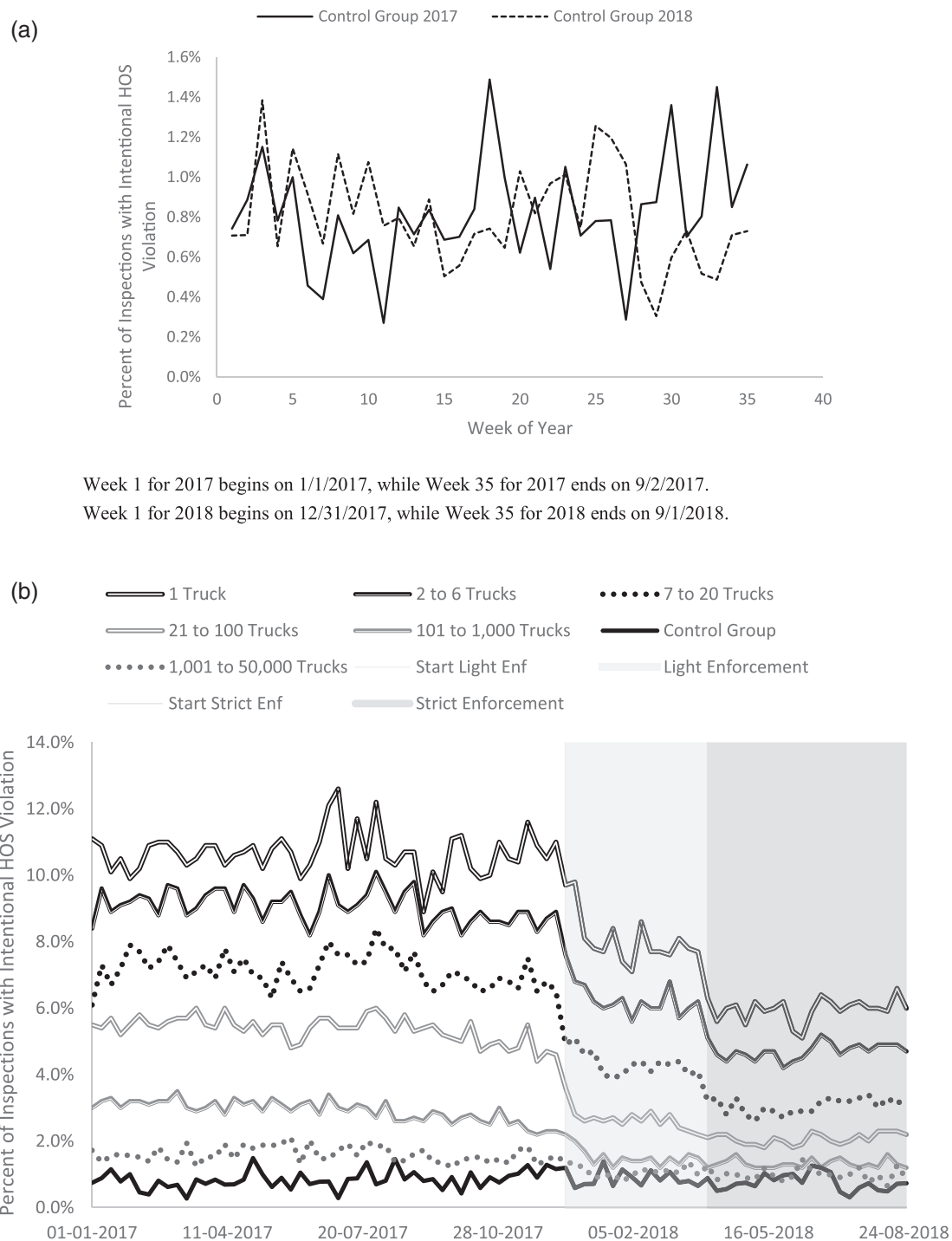


FIGURE 3 (a) Intentional HOS violation for the control group for common weeks. Week 1 for 2017 begins on 1/1/2017, while Week 35 for 2017 ends on 9/2/2017. Week 1 for 2018 begins on 12/31/2017, while Week 35 for 2018 ends on 9/1/2018. (b) Weekly intentional *HOS Violation Rates* by size cohort

especially for smaller carriers. Consequently, the available evidence suggests that the counterfactual assumptions underlying a DiD design for *HOS Violation Rate* are met.

Turning now to the results for *Log Unsafe Driving*, the only size cohort whose trend differs from the control group is the 1 truck cohort. As such, we can have good

confidence that the counterfactual assumptions for assigning causality for *Log Unsafe Driving* are met. Results concerning *Log Crashes* are more complex in that Model 5 indicates all size cohorts were trending upwards relative to our control group through August, but Model 6's results concerning the trend through the week prior to light enforcement of the ELD mandate indicates this

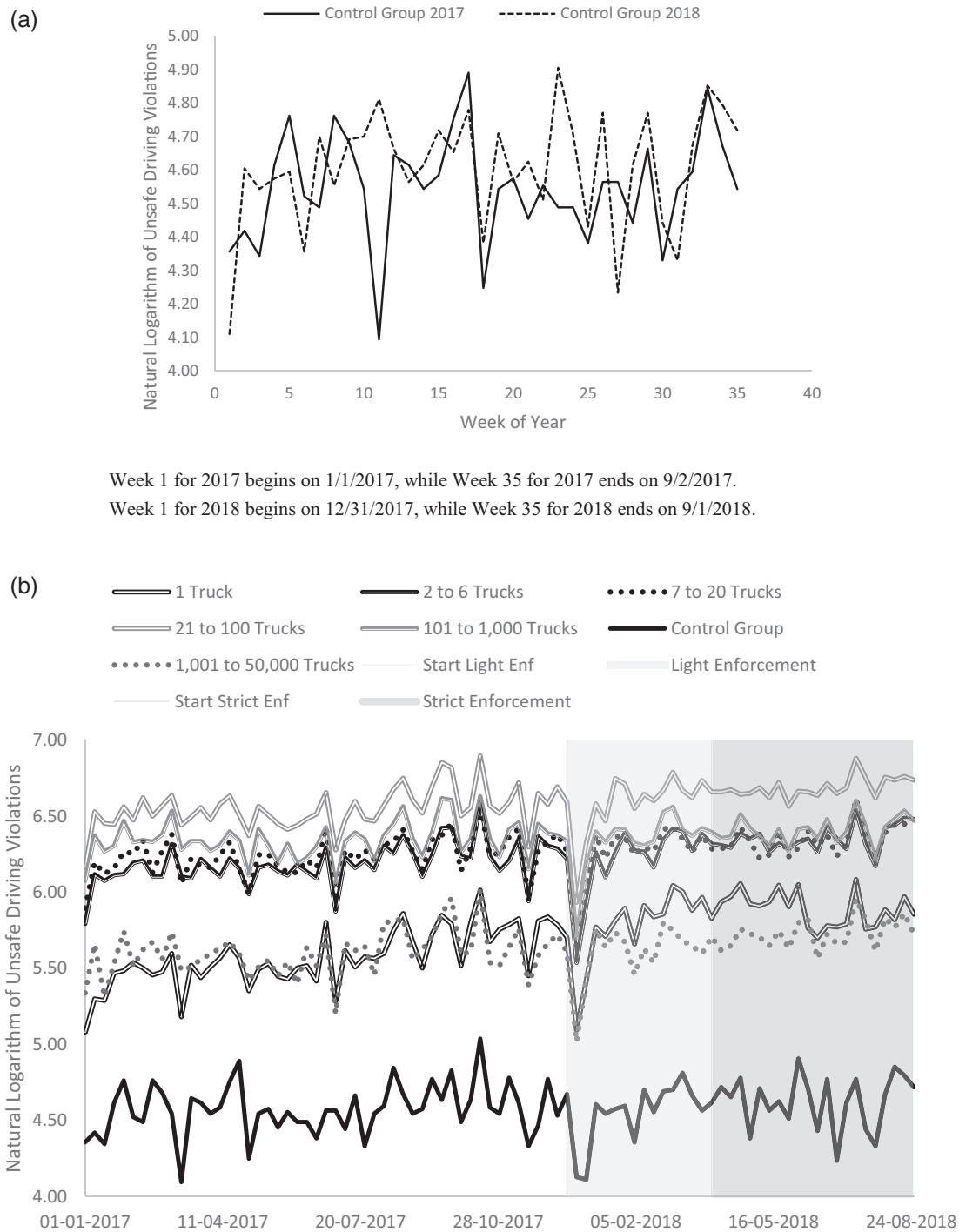


FIGURE 4 (a) Natural log of unsafe driving violations for the control group for common weeks. Week 1 for 2017 begins on 1/1/2017, while Week 35 for 2017 ends on 9/2/2017. Week 1 for 2018 begins on 12/31/2017, while Week 35 for 2018 ends on 9/1/2018. (b) Weekly natural log of unsafe driving violations by size cohort

upward trend was significant at $p < .05$ for the 1 truck and 2–6 truck size cohorts. One explanation is that crashes occur due to a large variety of stochastic factors beyond carriers' direct control (Corsi, Fanara, & Jarrell, 1988; Miller, Golcic, et al., 2018). As such, we believe readers should more cautiously interpret causal effects for *Log Crashes*.

5 | RESULTS

5.1 | Differences in means

The most transparent formulation of DiD estimation is to calculate the differences in the treatment mean before

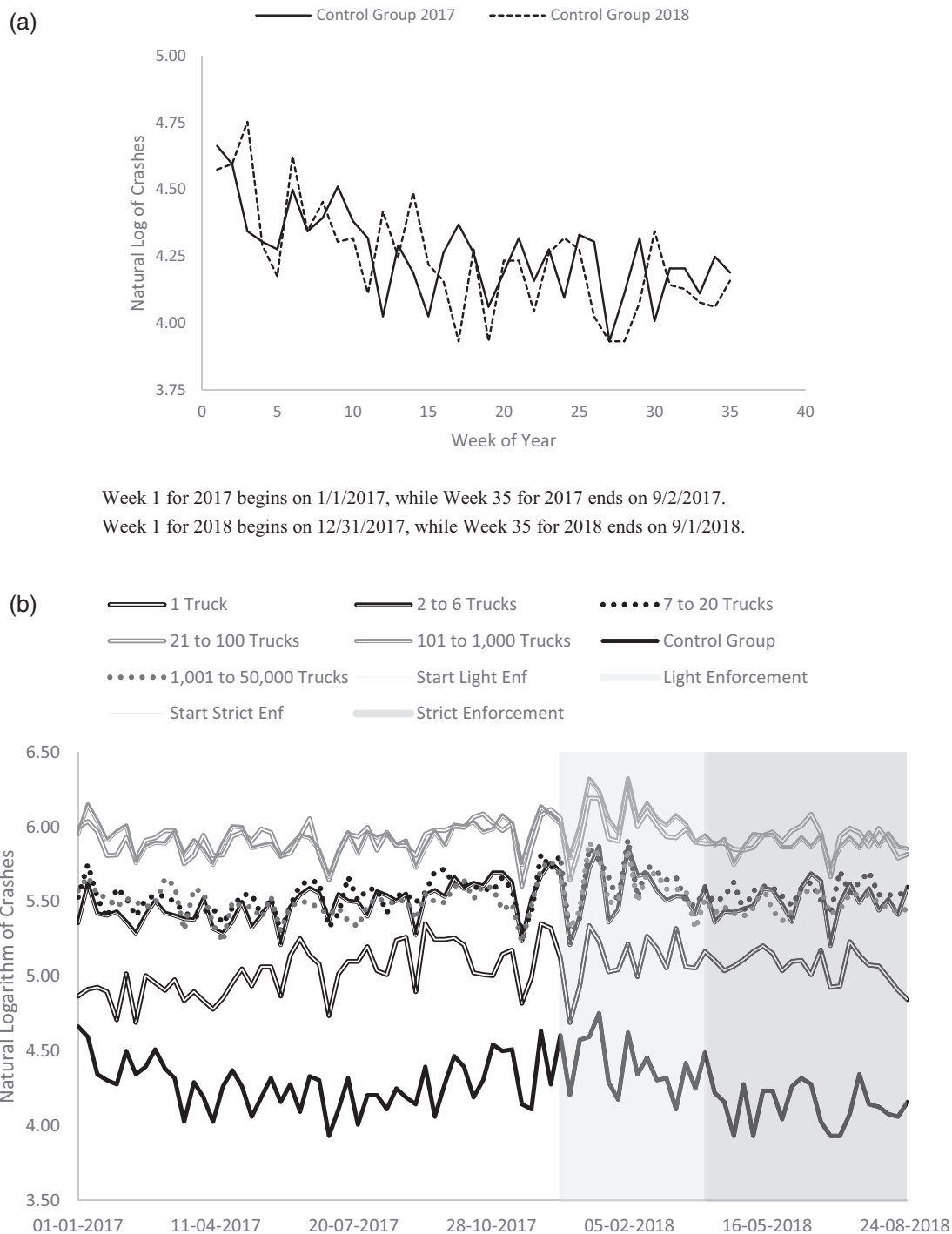


FIGURE 5 (a) Natural log of crashes for the control group for common weeks. Week 1 for 2017 begins on 1/1/2017, while Week 35 for 2017 ends on 9/2/2017. Week 1 for 2018 begins on December 31, 2017, while Week 35 for 2018 ends on 9/1/2018. (b) Weekly natural log of crashes by size cohort

and after the ELD mandate, and subtract from this the differences in the control group means. The means over different treatment periods, and their differences, can be found in Table 4.

Looking first at *HOS Violation Rate*, we see that violations for the control group are stable across the different

phases of the mandate implementation. HOS Violations tended to decrease for all treated groups, particularly for groups with fewer than 100 trucks, leading to negative DiD estimates, supporting H_{1A} . Also, the sizes of the DiD estimates are larger during strict enforcement than during light enforcement, which we would expect since

TABLE 3 Panel regression models to evaluate parallel trends

	Panel A: <i>HOS Violation Rate</i>		Panel B: <i>Log Unsafe Driving</i>		Panel C: <i>Log Crashes</i>	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Intercept	0.0069*** (5.89)	0.0070*** (8.21)	4.5079*** (62.29)	4.5052*** (81.36)	4.4205*** (72.75)	4.3232*** (61.09)
Linear trend	0.0001 (1.19)	0.0001* (1.84)	0.0021 (0.59)	0.0027 (1.43)	−0.0094*** (−3.19)	−0.0015 (−0.63)
1 Truck	0.0980*** (29.82)	0.0997*** (33.95)	0.8273*** (14.27)	0.8314*** (20.36)	0.3845*** (7.15)	0.5270*** (7.74)
2–6 Trucks	0.0840*** (35.92)	0.0863*** (48.48)	1.5593*** (21.99)	1.5676*** (31.52)	0.9489*** (16.18)	1.0390 (16.10)
7–20 Trucks	0.0629*** (21.72)	0.0659*** (28.68)	1.6355*** (24.55)	1.6443*** (36.21)	1.0668*** (19.49)	1.1474*** (18.40)
21–100 Trucks	0.0485*** (25.77)	0.0505*** (32.76)	1.94444 (29.30)	1.9462*** (41.25)	1.4786*** (27.67)	1.5541*** (25.53)
101–1,000 Trucks	0.0257*** (17.75)	0.0264*** (22.44)	1.7909*** (27.24)	1.7862*** (38.05)	1.5114*** (30.85)	1.5694*** (27.75)
1,001–50,000 Trucks	0.0086*** (5.56)	0.0097*** (8.05)	1.0034*** (15.01)	1.0129*** (21.42)	1.1011*** (19.88)	1.1565*** (19.14)
Linear trend × 1 truck	0.0001 (0.50)	−0.0001 (−0.58)	0.0071** (2.53)	0.0066*** (4.73)	0.0186*** (7.16)	0.0085*** (3.68)
Linear trend × 2–6 trucks	0.0000 (0.02)	−0.0002*** (−2.72)	0.0026 (0.76)	0.0018 (1.08)	0.0129*** (4.56)	0.0063*** (2.86)
Linear trend × 7–20 trucks	0.0001 (0.67)	−0.0001 (−1.58)	0.0016 (0.50)	0.0008 (0.51)	0.0097*** (3.68)	0.0040* (1.87)
Linear trend × 21–100 trucks	−0.0001 (−0.88)	−0.0002*** (−3.97)	0.0013 (0.40)	0.0011 (0.70)	0.0088*** (3.36)	0.0032 (1.55)
Linear trend × 101–1,000 trucks	−0.0002** (−2.57)	−0.0002*** (−5.58)	−0.0009 (−0.27)	−0.0007 (−0.41)	0.0072*** (3.02)	0.0027 (1.39)
Linear trend × 1,001–50,000 trucks	−0.0000 (−0.17)	−0.0001* (−1.95)	0.0013 (0.41)	0.0007 (0.44)	0.0064** (2.38)	0.0022 (1.08)
Observations	245	350	245	350	245	350
R ²	.980	.979	.981	.982	.977	.955

Note: Robust *t*-statistics reported in parentheses below parameter estimates. *SEs* calculated using the Parks (1967) method. Models 1, 3, and 5 represent trend projections through the end of August 2017 (Week 35), whereas Models 2, 4, and 6 represent trend projections through the week prior to the ELD mandate (Week 50).

**p* < .10.

***p* < .05.

****p* < .01 (two-tailed).

monitoring is more intense. Consistent with H_{1B} , the decrease in HOS violations is larger for size cohorts that had higher violation rates prior to the mandate.

For *Log Unsafe Driving*, the control group shows a U-shape pattern in violations over the different phases of treatment: it starts at a medium level, decreases during light enforcement, before increasing to a slightly higher than initial level. The treatment groups with 100 or fewer trucks show monotonic increases in violations. This results in DiD estimates that are positive during light enforcement, and relatively large during strict

enforcement, providing evidence in favor of H_{2A} . Furthermore, consistent with H_{2B} , the DiD estimates appear larger for the size cohorts for the smaller carriers which had higher rates of HOS violations prior to the light enforcement period.

Finally, for *Log Crashes* the control group shows an inverted-U pattern, with crashes starting at a medium level, rising during light enforcement, and then falling to a lower than initial level. Treatment groups display a similar inverted-U, but crashes never go below their initial level. The results are relatively close to 0 during light

TABLE 4 Differences-in-differences (DiD) tables for *HOS Violation Rate*, *Log Unsafe Driving*, and *Log Crashes*

A: HOS Violation Rate							
	1 Truck	2–6 Trucks	7–20 Trucks	21–100 Trucks	101–1,000 Trucks	1,001–50,000 Trucks	Control group
No enforcement	0.107	0.091	0.071	0.054	0.029	0.016	0.008
Light enforcement	0.081	0.063	0.043	0.027	0.015	0.011	0.009
Strict enforcement	0.060	0.047	0.031	0.021	0.013	0.010	0.008
Diff-in-diff light enforce	−0.026	−0.028	−0.028	−0.027	−0.014	−0.006	—
Diff-in-diff strict enforce	−0.046	−0.043	−0.040	−0.032	−0.015	−0.005	—
B: Log Unsafe Driving							
	1 Truck	2–6 Trucks	7–20 Trucks	21–100 Trucks	101–1,000 Trucks	1,001–50,000 Trucks	Control group
No enforcement	5.574	6.188	6.238	6.549	6.344	5.605	4.574
Light enforcement	5.771	6.214	6.261	6.577	6.334	5.597	4.551
Strict enforcement	5.887	6.355	6.348	6.689	6.408	5.728	4.628
Diff-in-diff light enforce	0.220	0.049	0.045	0.050	0.014	0.015	—
Diff-in-diff strict enforce	0.260	0.113	0.056	0.087	0.011	0.069	—
C: Log Crashes							
	1 Truck	2–6 Trucks	7–20 Trucks	21–100 Trucks	101–1,000 Trucks	1,001–50,000 Trucks	Control group
No enforcement	5.028	5.482	5.531	5.918	5.920	5.494	4.281
Light enforcement	5.104	5.567	5.628	5.995	6.065	5.594	4.401
Strict enforcement	5.071	5.502	5.562	5.908	5.888	5.480	4.148
Diff-in-diff light enforce	−0.044	−0.036	−0.023	−0.043	0.025	−0.020	—
Diff-in-diff strict enforce	0.177	0.153	0.165	0.124	0.102	0.120	—

enforcement, but large and positive during strict enforcement. This provides evidence for H_3 .

5.2 | Regression-based DiD specification

To implement the regression-based DiD approach (Angrist & Pischke, 2009; Ho, Lim, Reza, & Xia, 2017; Song, Tucker, & Murrell, 2015), we utilize the general specification shown in Equation (1).

$$y_{it} = \alpha + \gamma_i + \lambda_t + \sum_{i=1}^6 \delta_i \text{Size}_i \times D1_t + \sum_{i=1}^6 \beta_i \text{Size}_i \times D2_t + \mathbf{X}_{it} \boldsymbol{\theta} + \varepsilon_{it} \quad (1)$$

Equation (1) includes fixed effects for size cohort (γ_i) to control for time-invariant properties associated with

each size cohort, and week fixed effects (λ_t) to control for factors such as holiday weeks (and the associated increase or decrease in miles driven), average weather conditions, and the number of cars on the road. There are six size cohort fixed effects (treating the control group of large asset-based carriers as the reference) and 86 week fixed effects (treating the last week as the reference). The treatment effects of the light enforcement period for each size cohort relative to the control group are captured by the vector of coefficients (δ_i) for interaction terms⁶ $\text{Size}_i \times D1_t$. Size_i is equal to 1 for carrier size i , and 0 otherwise. $D1_t$ takes a value of 1 when week t falls within the light enforcement period (Weeks 51–65), and 0 otherwise. Similarly, the treatment effects of the strict enforcement period for each size cohort relative to the control group are captured by the vector β_i , with $\text{Size}_i \times D2_t$ interactions equal to 1 for carrier size i when week t falls within the strict enforcement period (Weeks 66–87), and 0 otherwise.

Estimating Equation (1) using OLS with no additional elements in $\mathbf{X}_{it}$ will closely resemble⁷ the DiD estimates reported in Table 4A–C. For robustness, and to increase the precision of our *SEs*, we estimate conditional DiD models that include *Inspection Proportion*.⁸ It is important to reiterate that this variable is the product of (a) the proportion of vehicles on the road of size class i , and (b) the propensity of inspectors to inspect that particular size class. If roadway composition is constant, then variation in *Inspection Proportion* controls for changes in inspector behavior. On the other hand, if the share of size class i on the road changes, while inspection behavior remains constant, then *Inspection Proportion* provides a measure of the proportion of size class i on the road. When *Inspection Proportion* is interpreted as a measure of roadway composition, then its inclusion will tend to understate the policy effect of the mandate because many small carriers threatened to discontinue service precisely because of the mandate. On the other hand, if inspector behavior changed as a result of the mandate, then the inclusion of *Inspection Proportion* is necessary so that the policy effect is not confounded by the monitoring effect. This is particularly important when *HOS Violation Rate* is the dependent variable; it is less important for *Log Unsafe Driving* and *Log Crashes*, where the decision to inspect is not a necessary condition for observation. Given there is no better variable for monitoring road composition, we are agnostic in our preference for conditioned or unconditioned results, leaving it to the reader's discretion.

5.3 | *HOS Violation Rate* results

As evident in Figure 3b and Table 4, *HOS Violation Rate* showed substantial declines for the size cohorts encompassing smaller carriers immediately following the start of light enforcement; further reductions are evident after strict enforcement commenced. Table 5 Model 1 contains the OLS estimates⁹ for the unconditional treatment effects using HCCME = 3 heteroskedastic consistent *SEs*¹⁰ per Long and Ervin (2000) using the PROC PANEL procedure in SAS Version 13.2. A Wald test for the set of δ_i terms is significant ($\Delta\chi^2 = 1,102.4$ with $df = 6$, $p < .01$) and the individual δ_i terms are all significant and negative, indicating that all size cohorts decreased their *HOS Violation Rates* following the start of light enforcement relative to the control group. Likewise, a Wald test for the set of β_i terms is significant ($\Delta\chi^2 = 4,852.0$ with $df = 6$, $p < .01$) and the individual β_i terms are all significant and negative, suggesting all size cohorts decreased their *HOS Violation Rates* more than the control group following the start of strict enforcement. Thus, the evidence supports

H_{1A} . Furthermore, consistent with strict enforcement further increasing electronic monitoring intensity, a Wald test of $\delta_1 = \beta_1$, $\delta_2 = \beta_2$, ..., $\delta_6 = \beta_6$ is significant ($\Delta\chi^2 = 465.1$ with $df = 6$, $p < .01$). Individual Wald tests for the equality of the treatment effects within each size cohort suggest that strict enforcement had a stronger effect for 1 truck, 2–6 truck, 7–20 truck, and 21–100 truck cohorts. Strict and light enforcement had similar effects for the larger cohorts with 101–1,000 trucks and 1,001–50,000 trucks, likely because the larger carriers had completed ELD adoption when light enforcement began (Miller et al., 2020). Consistent with H_{1B} , individual constraints for the treatment effects of the smaller carrier size cohorts (1 truck and 2–6 trucks) relative to the larger size cohorts (101–1,000 trucks and 1,001–50,000 trucks) for light enforcement (e.g., $\delta_1 = \delta_5$, $\delta_1 = \delta_6$, $\delta_2 = \delta_5$, $\delta_2 = \delta_6$) and strict enforcement (e.g., $\beta_1 = \beta_5$, $\beta_1 = \beta_6$, $\beta_2 = \beta_5$, $\beta_2 = \beta_6$) periods are all statistically significant at $p < .01$, indicating smaller carriers' were more intensely treated by the mandate in absolute terms. Finally, as seen in Model 2 where *Inspection Proportion* is included, there is no evidence that weeks in which a size cohort received a greater proportion of inspections affects the rate its carriers were cited for *HOS* violations. Coefficient estimates for the effect of light and strict enforcement are virtually identical across Model 1 and Model 2.

A last point worth noting is that the high R^2 values reported in Table 5 for this measure and the others (to be discussed) is not a methodological artifact. Rather, it stems from a combination of (a) the aggregation of safety events to the size cohort-week level canceling out idiosyncratic characteristics of any one safety event (Grunfeld & Griliches, 1960) and (b) the inclusion of size cohort and week fixed effects removing stable across size cohort variation and weekly variation that is homogeneous across size cohorts. The remaining variation stems from time-varying factors within a size cohort that exhibit heterogeneous responses across size cohorts for that week. As visible in Figures 3b, 4b, and 5b, this variation is small. With this being said, per Mauro (1990) and Frank (2000), high R^2 values greatly assuage concerns about endogeneity due to omitted variables.

5.4 | *Log Unsafe Driving* results

The DiD models for *Log Unsafe Driving* are reported in Table 5 Panel B, with Model 3 giving estimates from Equation (1) and Model 4 providing estimates with *Inspection Proportion* added. Prior to the mandate, many one truck operations threatened to stop driving or change occupations. If these operators were more likely to commit unsafe driving infractions, and they left en masse, we

TABLE 5 Differences-in-differences (DiD) results using two-way fixed effects OLS estimation

		Panel A: HOS Violation Rate		Panel B: Log Unsafe Driving		Panel C: Log Crashes	
	Label	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Light enforcement							
1 Truck	δ_1	−0.026*** (−13.69)	−0.027*** (−12.94)	0.220*** (5.89)	0.135*** (3.99)	−0.044 (−0.90)	−0.109** (−2.42)
2–6 Trucks	δ_2	−0.028*** (−20.66)	−0.028*** (−19.95)	0.049 (1.43)	0.104*** (3.14)	−0.036 (−0.82)	0.007 (0.16)
7–20 Trucks	δ_3	−0.028*** (−21.47)	−0.028*** (−20.54)	0.045 (1.42)	0.114*** (3.67)	−0.023 (−0.53)	0.030 (0.72)
21–100 Trucks	δ_4	−0.027*** (−21.89)	−0.027*** (−21.22)	0.050 (1.53)	0.110*** (3.38)	−0.043 (−1.07)	0.003 (0.08)
101–1,000 Trucks	δ_5	−0.014** (−11.83)	−0.014*** (−11.51)	0.014 (0.40)	0.041 (1.29)	0.025 (0.60)	0.046 (1.18)
1,001–50,000 Trucks	δ_6	−0.006*** (−4.35)	−0.006*** (−4.35)	0.015 (0.41)	−0.000 (−0.01)	−0.020 (−0.48)	−0.032 (−0.78)
Strict enforcement							
1 Truck	β_1	−0.046*** (−38.47)	−0.048*** (−35.01)	0.260*** (6.76)	0.131*** (3.73)	0.177*** (4.54)	0.078* (1.92)
2–6 Trucks	β_2	−0.043*** (−46.77)	−0.043*** (−46.58)	0.113*** (3.45)	0.127*** (3.90)	0.153*** (4.28)	0.164*** (4.67)
7–20 Trucks	β_3	−0.040*** (−40.62)	−0.040*** (−39.64)	0.056* (1.70)	0.098*** (3.07)	0.165*** (4.64)	0.197*** (5.49)
21–100 Trucks	β_4	−0.032*** (−36.09)	−0.032*** (−34.27)	0.087*** (2.63)	0.133*** (4.09)	0.124*** (3.65)	0.159*** (4.56)
101–1,000 Trucks	β_5	−0.015*** (−16.48)	−0.015*** (−16.50)	0.011 (0.34)	0.009 (0.29)	0.102*** (3.03)	0.100*** (2.94)
1,001–50,000 Trucks	β_6	−0.005*** (−5.67)	−0.006*** (−5.88)	0.069** (2.05)	0.023 (0.69)	0.120*** (3.53)	0.085** (2.46)
Size cohort fixed effects	γ_i	Included	Included	Included	Included	Included	Included
Week fixed effects	λ_t	Included	Included	Included	Included	Included	Included
Inspection proportion	θ		0.001 (1.47)		0.076*** (11.01)		0.058*** (6.75)
Observations		609	609	609	609	609	609
R ²		.991	.991	.989	.991	.980	.981

Note: Robust t -statistics reported in parentheses below parameter estimates. SEs calculated using the HCCME = 3 method.

* $p < .10$.

** $p < .05$.

*** $p < .01$ (two-tailed).

would expect unsafe violations to *decline* as a result of the mandate. Looking at Model 4 we see that *Inspection Proportion* is partially correlated with unsafe driving: when a size class makes up a higher share of what is inspected they are more likely to get unsafe driving violations.¹¹ Nonetheless, whether we include *Inspection Proportion* or not, we see that unsafe driving violations *increase* during strict enforcement for all carriers with fewer than 100 trucks relative to the control group. A Wald test for the vector of β_i terms in Model 4 is

significant ($\Delta\chi^2 = 107.5$ with $df = 6$, $p < .01$). Unsafe violations during the light enforcement period also become significant for all carriers with fewer than 100 trucks when *Inspection Proportion* is included. Likewise, a Wald test for the vector of δ_i terms is significant ($\Delta\chi^2 = 62.8$ with $df = 6$, $p < .01$). The magnitude of the coefficient estimates are substantial, with estimates ranging from a 7 to 26% increase in unsafe driving attributable to the mandate. Thus, H_{2a} is corroborated. However, there is limited evidence that the light and strict enforcement

treatment effects differed in that a Wald test of $\delta_1 = \beta_1$, $\delta_2 = \beta_2$, ..., $\delta_6 = \beta_6$ is marginally significant ($\Delta\chi^2 = 11.7$ with $df = 6$, $p < .10$), but individual Wald tests for the equality of the treatment effects within each size cohort are all non-significant. This may be due to the fact that the increase in electronic monitoring was more pronounced in the transition from pretreatment to light enforcement vis-à-vis from light enforcement to strict enforcement. However, consistent with H_{2B} , individual constraints for the treatment effects of the smaller carrier size cohorts (1 truck and 2–6 trucks) relative to the larger size cohorts (101–1,000 trucks and 1,001–50,000 trucks) for light enforcement (e.g., $\delta_1 = \delta_5$, $\delta_1 = \delta_6$, $\delta_2 = \delta_5$, $\delta_2 = \delta_6$) and strict enforcement (e.g., $\beta_1 = \beta_5$, $\beta_1 = \beta_6$, $\beta_2 = \beta_5$, $\beta_2 = \beta_6$) periods are all statistically significant at $p < .01$. This suggests offense displacement toward more unsafe driving was greater for smaller carriers than larger carriers, which is what we expected given smaller carriers were more likely to violate HOS rules prior to the ELD mandate.

5.5 | Log Crashes results

The DiD models for *Log Crashes* are reported in Table 5 Panel C, with Model 5 representing the unconditional model, and Model 6 providing conditional estimates with *Inspection Proportion*. *Inspection Proportion* is again significant and positive, indicating a one percentage point increase results in 5.8% more crashes, which is consistent with crashes stemming, in part, from having relatively more vehicles on the road. In the unconditioned Model 5, there is not significant evidence that crashes changed much in the light enforcement period relative to the pre-enforcement; nonetheless, during the strict enforcement phase, crashes increased for all size classes relative to the control group. Examining the conditioned results of Model 6, it appears light enforcement had the same limited impact on crashes for the treated size cohorts relative to the control group; the only significant coefficient is for the size cohort with 1 truck, and this estimate is negative. This can be partially explained by the sharp reduction of the relative share of the 1 truck size cohort during the first weeks of light enforcement evident in Figure 2b. If the more dangerous portion of 1 truck carriers were off the road initially, crash rates for the 1 truck cohort may have plausibly fallen. However, when these one truck carriers returned to the road in full force during the strict enforcement period, and when unsafe driving violations began to increase, crash rates did as well. Relative to our control group, all size cohorts saw a significant proportionate increase in crashes once strict enforcement went into effect vis-à-vis no enforcement. The magnitudes of

these effects are pronounced. For example, the size cohort for 7–20 trucks saw a 20% greater increase in crashes during strict enforcement relative to no enforcement vis-à-vis the control group. Furthermore, a Wald test of $\delta_1 = \beta_1$, $\delta_2 = \beta_2$, ..., $\delta_6 = \beta_6$ is significant ($\Delta\chi^2 = 44.6$ with $df = 6$, $p < .01$), and all individual Wald tests for the equality of the treatment effects within each size cohort all significant except for the size cohort with 101–1,000 trucks. These results align with H_3 .

5.6 | Robustness testing

We conducted three series of robustness tests. The first series involves re-estimating our models for *Log Unsafe Driving* using only violations for speeding. We conduct this test because speeding is a clear way to make up for lost productivity. In total, 70.5% of unsafe driving violations in the data are speeding related. We take the natural log of speeding violations at the size cohort level and use this measure, *Log Speeding*, as the dependent variable. We report these results in Table 6. To facilitate comparison, we replicate the *Log Unsafe Driving* results from Table 5. Looking at Table 6, we see near-identical conclusions. Furthermore, the estimates for *Log Speeding* are slightly smaller than those for *Log Unsafe Driving*. For the significant treatment effects when conditioning on *Inspection Proportion*, the *Log Speeding* results are 83.2% as large as those for *Log Unsafe Driving*, on average. This makes sense in that we would expect drivers trying to make up for lost productivity by complying with HOS rules would also engage in other types of unsafe behaviors outside of speeding.

The second series are “placebo” tests whereby we drop data after the start of the light enforcement period and examine whether treatment effects exist for a placebo intervention (St. Clair & Cook, 2015). The idea is that we should not observe significant treatment effects on dates where there was no change in policy (or ELD adoption). Insignificant estimates should lend credibility to our empirical methodology. We selected placebo enforcement dates of Week 36 (9/3/2017) and Week 41 (10/8/2017) to be close to when actual enforcement begins, but with sufficient data during the “treatment” period to allow for precise estimation. We report results in Table 7.

Beginning first with the models for *HOS Violation Rate* (Table 7 Panel A), we observe negative treatment effects. This is not surprising, given carriers tended to be forward-looking and started to adopt ELDs in anticipation of enforcement (Miller et al., 2020). However, what is critical to note is the magnitudes of the placebo effects from Table 7 Panel A are far smaller than the treatment effects for light enforcement and especially so for strict

TABLE 6 Differences-in-differences (DiD) robustness results using only speeding violations

		Panel A: Log Speeding		Panel B: Log Unsafe Driving	
	Label	Model 1	Model 2	Model 3	Model 4
Light enforcement					
1 Truck	δ_1	0.191*** (4.26)	0.112*** (2.65)	0.220*** (5.89)	0.135*** (3.99)
2–6 Trucks	δ_2	0.050 (1.21)	0.101** (2.40)	0.049 (1.43)	0.104*** (3.14)
7–20 Trucks	δ_3	0.034 (0.85)	0.097** (2.39)	0.045 (1.42)	0.114*** (3.67)
21–100 Trucks	δ_4	0.048 (1.15)	0.103** (2.41)	0.050 (1.53)	0.110*** (3.38)
101–1,000 Trucks	δ_5	0.012 (0.27)	0.037 (0.89)	0.014 (0.40)	0.041 (1.29)
1,001–50,000 Trucks	δ_6	−0.012 (−0.26)	−0.026 (−0.55)	0.015 (0.41)	−0.000 (−0.01)
Strict enforcement					
1 Truck	β_1	0.214*** (5.22)	0.095** (2.41)	0.260*** (6.76)	0.131*** (3.73)
2–6 Trucks	β_2	0.076** (2.18)	0.089** (2.58)	0.113*** (3.45)	0.127*** (3.90)
7–20 Trucks	β_3	0.025 (0.72)	0.064* (1.88)	0.056* (1.70)	0.098*** (3.07)
21–100 Trucks	β_4	0.077** (2.22)	0.120*** (3.48)	0.087*** (2.63)	0.133*** (4.09)
101–1,000 Trucks	β_5	−0.018 (−0.52)	−0.020 (−0.60)	0.011 (0.34)	0.009 (0.29)
1,001–50,000 Trucks	β_6	0.038 (1.03)	−0.004 (−0.12)	0.069** (2.05)	0.023 (0.69)
Size cohort fixed effects	γ_i	Included	Included	Included	Included
Week fixed effects	λ_t	Included	Included	Included	Included
Inspection proportion	θ		0.069** (9.21)		0.076*** (11.01)
Observations		609	609	609	609
R ²		.988	.990	.989	.991

Note: Robust *t*-statistics reported in parentheses below parameter estimates. SEs calculated using the HCCME = 3 method.

**p* < .10.

***p* < .05.

****p* < .01 (two-tailed).

enforcement reported in Table 5 Panel A. For example, the placebo effect for the 1 truck size cohort is −0.004, whereas the light enforcement treatment effect from Table 5 Model 2 is −0.027, which means the placebo treatment effect is only 14.8% the size of the light enforcement treatment effect. Similarly, the placebo treatment effect for the 7–20 truck size cohort is −0.006 from Table 7 Model 2, yet the light enforcement treatment effect from Table 5 Model 2 is −0.028, which means the placebo treatment effect is only 21.4% the size of the light

enforcement treatment effect. Thus, the fact the actual treatment effects are far larger than those from the placebo DiD models provides evidence that the start of enforcement of the ELD mandate is in fact what is driving the pronounced reductions in intentional HOS violations.

Turning now to the placebo results for *Log Unsafe Driving*, we find that only the 1 truck size cohort shows a significant placebo treatment effect. In contrast, all other size cohorts show a non-significant placebo treatment.

TABLE 7 Placebo test differences-in-differences (DiD) results

	Panel A: <i>HOS Violation Rate</i>		Panel B: <i>Log Unsafe Driving</i>		Panel C: <i>Log Crashes</i>	
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
<i>Placebo test</i>						
1 Truck	−0.005** (−2.17)	−0.004* (−1.85)	0.092** (2.56)	0.117*** (2.89)	0.037 (0.71)	−0.084 (−1.52)
2–6 Trucks	−0.006*** (−5.14)	−0.006*** (−5.40)	0.051 (1.44)	0.044 (1.10)	0.082* (1.90)	0.070 (1.41)
7–20 Trucks	−0.005*** (−4.54)	−0.006*** (−5.59)	0.027 (0.81)	0.037 (0.94)	0.050 (1.10)	−0.006 (0.12)
21–100 Trucks	−0.006*** (−4.75)	−0.008*** (−6.18)	0.036 (1.05)	0.018 (0.44)	0.015 (0.34)	−0.002 (−0.03)
101–1,000 Trucks	−0.005*** (−4.06)	−0.007*** (4.53)	0.005 (0.15)	0.003 (0.08)	0.016 (0.36)	0.035 (0.73)
1,001–50,000 Trucks	−0.003** (−2.23)	−0.003*** (−2.85)	0.021 (0.58)	−0.009 (−0.22)	−0.005 (−0.11)	−0.016 (−0.30)
Size cohort fixed effects	Included	Included	Included	Included	Included	Included
Week fixed effects	Included	Included	Included	Included	Included	Included
Inspection proportion	0.001 (1.43)	0.001 (1.15)	0.066*** (6.41)	0.061*** (5.99)	0.059*** (4.32)	0.077*** (5.52)
Observations	350	350	350	350	350	350
R^2	.991	.991	.991	.991	.979	.979

Note: Robust *t*-statistics reported in parentheses below parameter estimates. *SEs* calculated using the HCCME = 3 method. Models 1, 3, and 5 assume enforcement starts on Week 36 (9/3/2017), whereas Models 2, 4, and 6 assume enforcement starts on Week 41 (10/8/2017).

* $p < .10$.

** $p < .05$.

*** $p < .01$ (two-tailed).

Furthermore, the magnitude of the light enforcement treatment effects for the 2–6 truck and 7–20 truck size cohorts is far larger than the placebo treatments. For example, the placebo effect for the 2–6 truck size cohort is 0.044 from Table 7 Model 4, whereas the light enforcement treatment effect from Table 5 Model 4 is 0.104, which means the placebo treatment effect is only 42.3% the size of the light enforcement treatment effect. This provides some evidence of anticipation of enforcement on the part of one truck carriers; nonetheless, the major behavioral change regarding unsafe driving took place following the commencement of ELD enforcement. Turning lastly to the *Log Crashes* results, Table 7 Panel C suggests that the placebo treatment effects are nonsignificant for all size cohorts at $p < .05$. As with *Log Unsafe Driving*, this provides evidence that the change in behavior of the smaller size cohorts occurred following ELD enforcement.

Our third series of results evaluates whether our main results hold using alternative estimators. Since *HOS Violation Rate* is a fractional variable on a (0,1) sample space, we implemented Papke and Wooldridge's (1996) quasi-likelihood approach using the GLIMMIX procedure in SAS. These results are reported in Table 8 Panel

A. Examining these results, we reach the same conclusions, suggesting the results for *HOS Violation Rate* are not overly sensitive to model specification.

Given unsafe driving violations and crashes can be conceptualized as arising from counting processes whereby the occurrence of events should be reasonably independent, we fit overdispersed Poisson regression models as a robustness test (Gardner, Mulvey, & Shaw, 1995). We utilized overdispersed Poisson models rather than negative binomial models because the former displayed better fit per the AIC. Turning first to Panel B with the results for *Unsafe Driving*, we see the results almost perfectly mirror those reported by the OLS models using *Log Unsafe Driving*; this is not surprising, as the Poisson model utilizes a natural logarithm link function (Gardner et al., 1995). Likewise, the results in Panel C for *Crashes* are very similar those reported in Table 8 Panel C.

5.7 | Potential threats to inferences

Given that our findings call into question the efficacy of a government intervention that has profound implications

TABLE 8 Robustness testing results using alternative estimators

	Label	Panel A: <i>HOS Violation Rate</i>		Panel B: <i>Unsafe Driving</i>		Panel C: <i>Crashes</i>	
		Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
<i>Light enforcement</i>							
1 Truck	δ_1	−0.353*** (−5.90)	−0.373*** (−6.19)	0.224*** (5.38)	0.150*** (3.94)	−0.049 (−1.06)	−0.100** (−2.20)
2–6 Trucks	δ_2	−0.442*** (−7.31)	−0.428*** (−7.06)	0.057 (1.45)	0.100*** (2.79)	−0.031 (−0.71)	0.001 (0.01)
7–20 Trucks	δ_3	−0.571*** (−9.23)	−0.554*** (−8.91)	0.049 (1.25)	0.103*** (2.86)	−0.018 (−0.41)	0.022 (0.51)
21–100 Trucks	δ_4	−0.773*** (−11.98)	−0.758*** (−11.73)	0.056 (1.45)	0.103*** (2.93)	−0.040 (−0.97)	−0.006 (−0.14)
101–1,000 Trucks	δ_5	−0.704*** (−10.10)	−0.697*** (−10.02)	0.018 (0.47)	0.040 (1.11)	0.026 (0.63)	0.042 (1.04)
1,001–50,000 Trucks	δ_6	−0.439*** (−5.86)	−0.443*** (−5.93)	0.015 (0.35)	0.001 (0.00)	−0.015 (−0.35)	−0.024 (−0.57)
<i>Strict enforcement</i>							
1 Truck	β_1	−0.530*** (−9.64)	−0.562*** (−9.90)	0.249*** (7.07)	0.145*** (4.37)	0.171*** (3.98)	0.094** (2.15)
2–6 Trucks	β_2	−0.602*** (−10.80)	−0.598*** (−10.78)	0.108*** (3.22)	0.120*** (3.96)	0.154*** (3.80)	0.162*** (4.13)
7–20 Trucks	β_3	−0.786*** (−13.71)	−0.775*** (−13.53)	0.052 (1.56)	0.087*** (2.86)	0.164*** (4.06)	0.189*** (4.80)
21–100 Trucks	β_4	−0.886*** (−14.88)	−0.875*** (−14.69)	0.080** (2.44)	0.118*** (3.96)	0.125*** (3.20)	0.153*** (4.00)
101–1,000 Trucks	β_5	−0.709*** (−11.15)	−0.710*** (−11.20)	0.007 (0.22)	0.004 (0.14)	0.102*** (2.60)	0.100*** (2.64)
1,001–50,000 Trucks	β_6	−0.382*** (−5.63)	−0.393*** (−5.81)	0.061* (1.72)	0.024 (0.73)	0.119*** (2.93)	0.091** (2.30)
Size cohort fixed effects	γ_i	Included	Included	Included	Included	Included	Included
Week fixed effects	λ_t	Included	Included	Included	Included	Included	Included
Inspection proportion	θ		0.019** (2.14)		0.062*** (10.74)		0.045*** (5.79)
Pearson χ^2/df		—	—	1.475	1.202	1.316	1.237
Observations		609	609	609	609	609	609
−2 log likelihood		203.5	203.5	5,461.2	5,322.7	5,065.1	5,023.6
R ²		.993	.993	.990	.991	.979	.980

Note: t-statistics reported in parentheses below parameter estimates. *HOS Violation Rate* models are estimated using Papke and Wooldrige's (1996) quasi-likelihood approach for fractional data. Unsafe Driving Violation and Crashes models are estimated using overdispersed Poisson regression models. R^2 values are calculated by outputting predicted values, applying an inverse link transformation, and correlating the transformed values with the raw data.

* $p < .10$.

** $p < .05$.

*** $p < .01$ (two-tailed).

on the largest transportation sector in the United States, we want to address other potential concerns about possible threats to the validity of our inferences. One potential concern is that inspectors' behaviors changed following the ELD mandate. For example, if inspectors began to

assume that if drivers had ELDs that they were operating within regulations, we could see a drop in *HOS Violation Rates*. This possibility, however, appears untenable in that the rate of violations for the control group remained unchanged during the light and strict enforcement

periods relative to no enforcement (see Table 4A). It seems highly unlikely that inspectors would systematically begin to less strictly enforce HOS regulations for smaller carriers that have historically had worse compliance with HOS rules (Miller, Golobic, et al., 2018) than the large carriers in the control group that have historically had low rates of HOS violations. Additionally, as shown in Figure 2a, the number of inspections remained practically unchanged in 2017 and 2018; this removes concerns that inspectors were performing their jobs in a radically different manner. As such, it seems a stretch to attribute our findings to changes in enforcement.

Another potential threat to inference is that the observed increases in unsafe driving behaviors and crashes for the smaller size cohorts relative to the control group stems from these size cohorts having grown in size during our time window. If this were the case, these size cohorts could have been more exposed to enforcement and especially crashes. Regarding this concern, we make three observations. First, our treatment effects remain even after including *Inspection Proportion* as a control, which is the best way to proxy for a size cohort's presence on the roads during a given week. Second, the pronounced increases in *Log Unsafe Driving* exhibited by the smaller size cohorts cannot be accounted for by growth alone. Using the raw counts of violations, for the size cohort for carriers with one truck, unsafe driving

violations increased by 22.7% during light enforcement and 35.3% during strict enforcement relative to no enforcement. Likewise, once strict enforcement went into effect, unsafe driving violations increased by 17.5, 11.1, and 14.2% for the size cohorts for 2–6 trucks, 7–20 trucks, and 21–100 trucks, respectively. However, the Bureau of Labor Statistics reports that the entire NAICS 484 sector (which encapsulates trucking) grew by only 3.3% from January 2017 through August 2018. Thus, even if one were to argue that exposure to unsafe driving violations increases for a size cohort as more trucks are added, these increases are disproportional. Third, it is critical to remember that our vector of week fixed effects accounts for any sectoral trends that have the same effects across size cohorts. Consequently, and particularly with regard to unsafe driving violations, it does not seem reasonable to attribute our findings to size cohort differences in growth rates.

A third potential concern, as we noted in §4.2, is that we cannot directly observe ELD adoption. In this regard, we obtained inspection data starting during the light enforcement period and running through strict enforcement for the percentage of inspections where carriers were cited for not having ELDs installed; these violations are *not* included in the current intentional HOS violation list. In Figure 6 we plot the data over the time period during our study. A dramatic drop can be observed right as

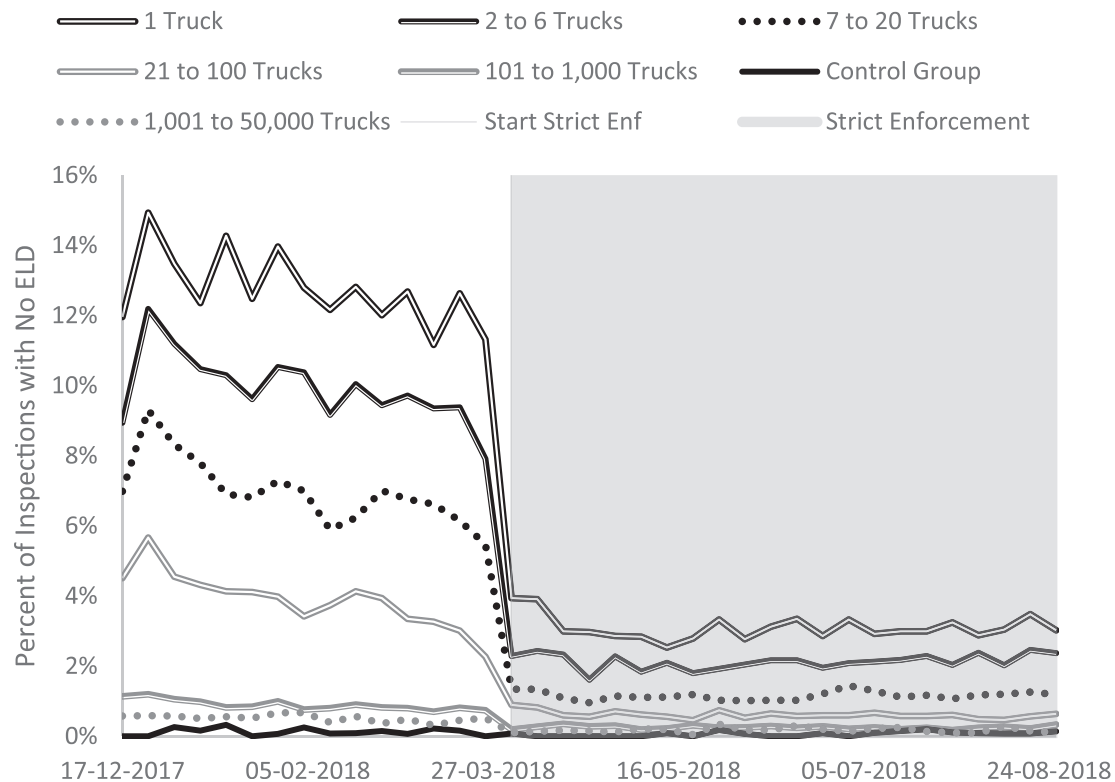


FIGURE 6 Plot of carriers' rate of being cited for not having ELDs installed. ELDs, electronic logging devices

the transition from light to strict enforcement takes place, which is consistent with our arguments that enforcement of the ELD is what is causing carriers to utilize ELDs. Furthermore, Figure 6 indicates that about 7% of inspections for the size cohort for carriers with 7–20 trucks and 4% of inspections for the size cohort for carriers with 21–100 trucks resulted in violations for not having ELDs during the light enforcement period. However, just 3 months prior to this, Miller et al. (2020) report that only 15% of carriers with 5–99 power units had started the process of installing ELDs.

6 | DISCUSSION

6.1 | Theoretical contributions

To date, existing research has found that increased monitoring of an agent's behavior reduces agents' propensity to commit illicit acts (Banerjee et al., 2008; Duflo et al., 2012; Nagin et al., 2002). Furthermore, research suggests the existence of positive spillover effects (i.e., positive second-order effects) from increased monitoring onto performance dimensions outside the direct purview of monitoring (Pierce et al., 2015). Our work contributes to the literature by drawing on the concept of offense displacement (Guerette & Bowers, 2009) to examine whether greater monitoring on one dimension (HOS compliance) can result in agents shifting the type of illicit behavior to another dimension (e.g., unsafe driving) that is not subject to increased monitoring (Yang, 2008).

This raises the question: why does offense displacement occur in some settings but not others? We believe the answer stems from important contextual factors in each particular industry. For example, in Pierce et al.'s (2015) restaurant context, once the IT monitoring system curtailed gains via theft, restaurant staff had little recourse for increasing earnings apart from improving service (resulting in higher tip percentages) and working harder to sell more food and drinks (resulting in higher tips). There was simply no outlet for offense displacement because both sets of behaviors were desirable for restaurant owners (principals). In contrast, requiring truck drivers to track their hours using ELDs did not curtail their ability to increase their pace of work (e.g., speeding). Moreover, since enforcement for unsafe driving tends to be perceived as limited or easy to circumvent (Levy, 2015; Ouellet, 1994), it seems likely that many truckers would seek to make up for lost productivity by increasing the pace that they work by speeding and driving more aggressively.

A second way this manuscript contributes to theory is by suggesting that the degree of offense displacement is moderated by the extent that agents committed the now-

monitored illicit act prior to monitoring. Specifically, we draw on the tax evasion literature (Kleven et al., 2011; Slemrod et al., 2001), which has proposed that agents who are unlikely to commit the illicit act that is to be targeted for greater monitoring are unlikely to be affected by changing monitoring regimes (Pomeranz, 2015). Consequently, these agents will have little incentive to engage in offense displacement. In contrast, agents that were more likely to commit the now-monitored illicit act are more likely to engage in offense displacement to other illicit acts to make up for lost utility (Yang, 2008). Our empirical results corroborate this prediction in that smaller carriers that had higher rates of HOS violations prior to the ELD mandate saw a greater increase in unsafe driving violations following the start of the ELD mandate than larger carriers that had lower rates of HOS violations prior to the ELD mandate. This is theoretically important in that it suggests a key boundary condition for which agents are most likely to engage in offense displacement.

A third way this manuscript contributes to theory is by challenging the prevailing wisdom from prior motor carrier research that higher levels of electronic monitoring, be it through ELDs or more comprehensive systems incorporating ELDs, result in reduced unsafe driving behaviors (Hickman & Hanowski, 2011) or fewer crashes by reducing HOS violations (Cantor et al., 2009). This study goes beyond Miller, Saldanha, et al.'s (2018) caution that greater use of electronic monitoring will show diminishing returns with regard to safety by suggesting that requiring the vast majority of commercial trucks be equipped with ELDs had little discernible effect on safety, and may have made U.S. roads less safe.

6.2 | Managerial and policy implications

The results from this study have implications for various stakeholders. Beginning with operations managers, our study highlights that the greater use of electronic monitoring to target one subset of agents' behaviors can have complex effects that can call into question the overall efficacy of heightened electronic monitoring. For example, while we found evidence consistent with prior research that increased monitoring to reduce agents' propensity to commit one form of illicit behavior reduces the frequency of said behavior (Pierce et al., 2015; Staats et al., 2017), we provide evidence that agents may increase their propensity to commit another form of illicit act. To the extent that the offense displacement behaviors may have a greater negative marginal effect on principals' outcomes (Guerette and Bowers (2009), our results suggest that increased monitoring may not only fail to improve principals' outcomes, it could actually make these outcomes

worse. Thus, we urge operations managers to fully consider whether greater monitoring efforts to reduce one type of illicit worker behavior is likely to result in an increase in other types of illicit behaviors that can be even more counterproductive.

This raises the questions: are there certain industry settings where offense displacement is likely to be more problematic? Our read of the literature suggests this issue is more likely in settings where agents have greater autonomy in determining how delegated tasks are executed. Thus, we would expect offense displacement due to greater monitoring to be more likely to occur in remote repair crews than, say, warehouse workers (Zuboff, 1988). The reason is straightforward: more agent autonomy creates more dimensions along which agents can engage in other undesired actions. A second dimension concerns the extent agents have a collective negative perception regarding greater monitoring because it is viewed as coercive (Sewell & Barker, 2006). For example, the increase in invisibility practices (e.g., agents taking longer breaks than allowed) by TSA workers at a large airport in response to heightened electronic surveillance was, in part, facilitated by workers collectively perceiving the surveillance as coercive (Anteby & Chan, 2018). We would expect collective perceptions to be stronger in unionized settings (Roscigno & Hodson, 2004) as well as settings where agents frequently interact with one another (Roy, 1952). For example, offense displacement may be more likely to occur in factories, where workers on the same shift tend to take breaks at the same time, relative to retail sales clerks, who take breaks on different schedules. A third factor is the extent principals are able to utilize the information gleaned from greater monitoring in a manner that agents view as beneficial, such as identifying free-riders (Sewell, Barker, & Nyberg, 2012). To the extent that principals can leverage greater information from monitoring to take actions that are perceived as benefitting agents, we would expect offense displacement behaviors to be less frequent.

Another implication for operations managers is that agents who were more likely to commit a now-monitored illicit act will be those who are most likely to display offense displacement to other illicit acts (Guerette & Bowers, 2009). As such, our results can help operations managers pinpoint which individuals are most likely to transition to other types of illicit acts and adjust other managerial control levers accordingly. In particular, our results suggest that offense displacement is of limited concern for individuals who were unlikely to commit the now-monitored illicit act in the first place.

We now turn to implications for industry-specific stakeholders including carriers, shippers, insurance companies, the FMCSA, and policy makers. Beginning first

with carriers, our findings suggest that carrier managers should be on the increased lookout for drivers who are accruing unsafe driving violations, especially at smaller-sized carriers. As explained by Straight (2018), such violations are problematic because they not only count against carriers' publicly available safety scores, they further increase insurance costs. With regard to shippers, our findings suggest that they should closely examine whether the carriers with whom they have long-term contracts have seen unsafe driving violations and crashes increase in frequency since the ELD mandate took effect, with special emphasis placed on smaller carriers. To the extent that shippers interpret such increases as a lack of sound operating routines and poor managerial control (Lehman, Cooil, & Ramanujam, 2019), they may want to reduce their tender offerings (Scott, 2019) to these carriers and not invite these carriers when they conduct their next strategic procurement auction (Caplice, 2007). Similarly, insurance companies should closely examine whether the carriers they insure have seen an increase in unsafe driving violations and crashes and adjust rates accordingly.

Turning now to regulators and policy makers, one implication from our findings is that, due to smaller carriers increasing their number of unsafe driving violations—the majority of which are speeding related—serious consideration should be put toward fully understanding the interaction between multiple bright-line rules and the mechanisms to enforce these rules. Bright-line rules refer to clearly specified standards that limit agents' discretion (Glaeser & Shleifer, 2001), such as HOS rules and speed limits. By making HOS rules more binding, the ELD mandate had the unintended consequence of causing truckers who were more likely to violate HOS rules prior to the ELD mandate subsequently increasing their unsafe driving behaviors by a greater degree once the mandate took effect. This is analogous to teachers in low-performing classrooms being more likely to falsify standardized test results once their incentives became tied to students' test scores reaching target thresholds (Jacob & Levitt, 2003). A second implication for regulators is that, if their intent is to get *HOS Violation Rates* to be universally low to the level exhibited by the control group, then the ELD mandate has not fully achieved its intended first-order goals per Figure 3b. As such, it appears that further changes in enforcement would be necessary to bring the intentional *HOS Violation Rates* for the smaller size cohorts down to the level exhibited by the largest carriers.

6.3 | Limitations

This study has several limitations. First, we cannot directly observe carriers' adoption of ELDs; rather, we

infer that the FMCSA beginning the different enforcement regimes is what drives ELD adoption. While the precipitous drops observed in Figure 3b in the first weeks where the ELD mandate is enforced, coupled with the very rapid decline in the number of inspections where carriers were cited for not having ELDs once strict enforcement took effect (Figure 6), are consistent with our arguments, this is a limitation nonetheless. However, our estimated treatment effects are conservative because some smaller carriers utilized ELDs well before the mandate (Miller et al., 2020). A second limitation is that we cannot measure different size cohorts' levels of exposure to inspectors at a weekly level. While we have sought to control for this as best as possible using the proportion of total weekly inspections each size cohort represents, this is admittedly an imperfect proxy. A third limitation stems from the fact our study focuses on a single industry: the for-hire, interstate U.S. motor carrier sector. As noted by Joglekar, Davies, and Anderson (2016), while single industry studies are of immense practical value, their conclusions must be generalized cautiously to other sectors. For example, the fact that truck drivers are fiercely independent and resistant to new regulations (Levy, 2015) may make offense displacement more likely.

6.4 | Directions for future research

This research suggests several directions for further research. One direction is for scholars to examine broader long-term consequences of the ELD mandate. For example, since trucks equipped with engines predating 2000 are exempt from the ELD Mandate (Meer, 2017), it would be interesting to see if the ELD mandate has resulted in a shift toward some carriers utilizing older vehicles. Given older equipment is less fuel efficient and requires more maintenance, such actions could lower the quality of equipment operating on U.S. roadways. Second, it would be worthwhile to examine whether the ELD mandate ultimately results in a heightened number of carrier failures. Third, it would be interesting to identify whether carriers that were late to comply with the ELD mandate have different longitudinal safety trajectories (Miller, Fugate, & Golicic, 2017) post-enforcement vis-à-vis carriers that were early compliers. Fourth, the fact some states gave inspectors leeway regarding whether they enforced the ELD mandate during the light enforcement period (Dills, 2018) provides a natural experiment that would allow scholars to explore heterogeneity in individual inspectors' behaviors, a topic that has seen limited attention (Becker, Jin, & Leslie, 2012; Feinstein, 1989; Macher, Mayo, & Nickerson, 2011; Scott, 2018). For example, scholars could explore how inspectors' prior

stringency or other characteristics that can be derived from the inspection file affected whether or not they cited carriers for not having ELDs during the light enforcement period.

6.5 | Conclusion

Researchers have found that greater monitoring of agents' (e.g., workers) behaviors improves agent behavior along the monitored dimensions, such as reduced absenteeism (Banerjee et al., 2008), less shirking (Nagin et al., 2002), increased compliance with standardized processes (Staats et al., 2017), less fraud (Pomeranz, 2015) and less theft (Pierce et al., 2015). We extend this work by describing how greater use of information technologies to monitor agents can result in negative spillovers and unintended consequences. We examine this question using a DiD design to study how truck drivers' safety behaviors and safety outcomes changed following a regulatory mandate that required most large commercial trucks install ELDs to record truckers' HOS. While we find that the ELD mandate reduced truckers' propensity to violate HOS rules, we also find that smaller firms—which had very low rates of ELD adoption prior to the regulatory mandate—saw a substantial uptick in their citations for unsafe driving behaviors. We further find that relative to our control group, all size cohorts saw significant increases in their crash rates once strict enforcement of the ELD mandate started. These results call into question whether increased electronic monitoring has improved safety in this industry sector and suggest key contingencies managers and policy makers should consider when designing and implementing programs to monitor agents with information technologies.

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ENDNOTES

¹ To enforce HOS regulations, the FMCSA requires drivers to log their work schedules, and DOT-approved inspectors check these logs periodically via unannounced roadside inspections (Cantor, Corsi, Grimm, & Singh, 2016). During a roadside inspection, an inspector checks a driver's documents—medical certificates, licenses, and electronic or paper logs—to ensure that they are operating legally. The inspector can then end the inspection or perform a more thorough vehicle inspection (Kahaner, 2015).

² Some carriers, especially those with 50–100 trucks, used an alternative technology called automatic on-board recording devices (AOBRDs) prior to the ELD mandate (Gallagher, 2019). Carriers using AOBRDs were allowed to delay ELD adoption until December 17, 2019. While AOBRDs are not as restrictive as ELDs

(FMCSA, 2019), AOBRDs provide more detail about drivers' work hours than paper logbooks. Thus, to the extent carriers in our treatment groups used AOBRDs and did not transition to ELDs, our treatment effects will be under-estimated because these carriers did not increase their electronic monitoring intensity.

- ³ Data can be purchased here: https://ask.fmcsa.dot.gov/app/mcmiscatalog/c_chap4. We are happy to share any and all of our data with interested researchers except the source files, which are purchased from the FMCSA.
- ⁴ These carriers are Knight Transportation, Swift Transportation, Werner Enterprises, Schneider National, U.S. Xpress, CR England, JB Hunt Transportation, and Maverick Transportation
- ⁵ Due to the use of size cohorts, what matters is the total number of trucks within a size cohort, not the number of unique carriers within a cohort.
- ⁶ Since all carriers are subject to ELD enforcement, the treatment dummies $D1_t$ and $D2_t$ are indexed by t , not i and t . Consequently, the linear terms for $D1_t$ and $D2_t$ cannot be included in Equation (1) because they are perfectly correlated with weekly time dummies λ_t .
- ⁷ They do not exactly replicate Table 4A–C because Equation (1) includes week fixed effects.
- ⁸ To make this parameter more interpretable, we multiply the proportion by 100 prior to including it in the models.
- ⁹ One issue readers may have is that *HOS Violation Rate* lies on a (0,1) continuum, which can raise concerns about the use of an OLS estimator due to predictions beyond the range of the data (Ramalho, Ramalho, & Murteira, 2011). We utilize an OLS estimator for four reasons. First, no predicted values lie beyond the range of the data. Second, we are interested in the absolute reduction in *HOS Violation Rate*, which is best estimated using an identity link function. In contrast, the logistic link function utilized for fractional regression models estimates parameters capturing proportional reductions. Third, the OLS estimator returns direct estimates of average treatment effects; the same is not true when using a nonlinear link function. Fourth, as a robustness test we utilize the quasi-likelihood approach suggested by Papke and Wooldridge (1996) and reach the same conclusions.
- ¹⁰ We use the HCCME = 3 approach since it is the preferred technique for calculating heteroskedastic-consistent *SEs* per Long and Ervin (2000). Angrist and Pischke (2009, p. 300) note these *SEs* tend to be the most conservative. Additionally, the HCCME = 3 *SEs* are larger than conventional *SEs*; thus, using the HCCME = 3 *SEs* aligns with Angrist and Pischke's (2009, p. 307) recommendation to use the maximum of the conventional *SE* and robust *SE*. Furthermore, we do not cluster errors by size cohorts because there are only seven cohorts, well under the recommended number of groups suggested by Angrist and Pischke (2009). Also, when we did try to cluster *SEs* and corrected for heteroskedasticity, the new estimates were ~40% smaller than those obtained from just correcting for heteroskedasticity.
- ¹¹ Note that the interpretation of *Inspection Proportion* changes in 5.3 (and 5.4) relative to 5.2. Because unsafe driving violations are moving violations, bias in inspector behavior at random roadside checks are less of a concern. Trucks are pulled over primarily because the law enforcement officer has witnessed some

infraction. We thus use *Inspection Proportion* as a measure of what class of vehicles is on the road in any given week.

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